

Problems

Section 6.1

6.1 The current in a $150\ \mu\text{H}$ inductor is known to be

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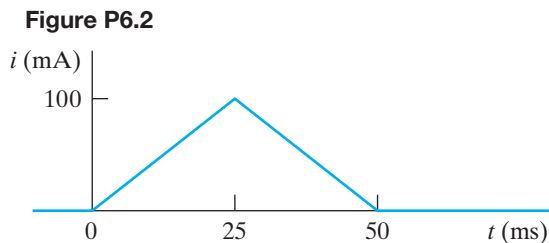
$$i_L = 25te^{-500t}\ \text{A} \quad \text{for } t \geq 0.$$

- Find the voltage across the inductor for $t > 0$. (Assume the passive sign convention.)
- Find the power (in microwatts) at the terminals of the inductor when $t = 5\ \text{ms}$.
- Is the inductor absorbing or delivering power at $5\ \text{ms}$?
- Find the energy (in microjoules) stored in the inductor at $5\ \text{ms}$.
- Find the maximum energy (in microjoules) stored in the inductor and the time (in milliseconds) when it occurs.

6.2 The triangular current pulse shown in Fig. P6.2 is applied to a $500\ \text{mH}$ inductor.

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- Write the expressions that describe $i(t)$ in the four intervals $t < 0$, $0 \leq t \leq 25\ \text{ms}$, $25\ \text{ms} \leq t \leq 50\ \text{ms}$, and $t > 50\ \text{ms}$.
- Derive the expressions for the inductor voltage, power, and energy. Use the passive sign convention.



6.3 The current in a $50\ \text{mH}$ inductor is known to be

$$i = 120\ \text{mA}, \quad t \leq 0;$$

$$i = A_1e^{-500t} + A_2e^{-2000t}\ \text{A}, \quad t \geq 0.$$

The voltage across the inductor (passive sign convention) is $3\ \text{V}$ at $t = 0$.

- Find the expression for the voltage across the inductor for $t > 0$.
- Find the time, greater than zero, when the power at the terminals of the inductor is zero.

6.4 Assume in Problem 6.3 that the value of the voltage across the inductor at $t = 0$ is $-18\ \text{V}$ instead of $3\ \text{V}$.

- Find the numerical expressions for i and v for $t \geq 0$.
- Specify the time intervals when the inductor is storing energy and the time intervals when the inductor is delivering energy.

c) Show that the total energy extracted from the inductor is equal to the total energy stored.

6.5 The current in a $300\ \text{mH}$ inductor is

$$i = 100\ \text{mA}, \quad t \leq 0;$$

$$i = (B_1 \cos 220t + B_2 \sin 220t)e^{-60t}\ \text{A}, \quad t \geq 0.$$

The voltage across the inductor (passive sign convention) is $5\ \text{V}$ at $t = 0$. Calculate the power at the terminals of the inductor at $t = 30\ \text{ms}$. State whether the inductor is absorbing or delivering power.

6.6 Evaluate the integral

$$\int_0^{\infty} p\ dt$$

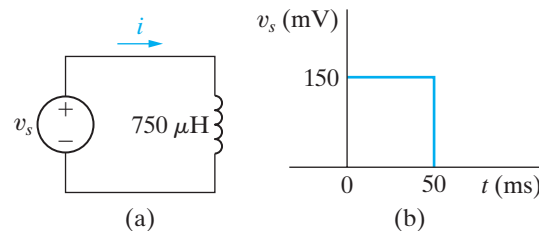
for Example 6.2. Comment on the significance of the result.

6.7 The voltage at the terminals of the $750\ \mu\text{H}$ inductor in Fig. P6.7(a) is shown in Fig. P6.7(b). The inductor current i is known to be zero for $t \leq 0$.

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- Derive the expressions for i for $t \geq 0$.
- Sketch i versus t for $0 \leq t \leq \infty$.

Figure P6.7



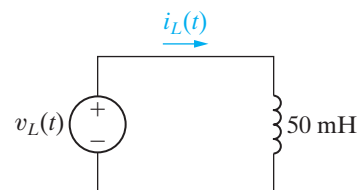
6.8 The current in the $50\ \text{mH}$ inductor in Fig. P6.8 is known to be $100\ \text{mA}$ for $t < 0$. The inductor voltage for $t \geq 0$ is given by the expression

$$v_L(t) = 2e^{-100t}\ \text{V}, \quad 0^+ \leq t \leq 100\ \text{ms}$$

$$v_L(t) = -2e^{-100(t-0.1)}\ \text{V}, \quad 100\ \text{ms} \leq t < \infty$$

Sketch $v_L(t)$ and $i_L(t)$ for $0 \leq t < \infty$.

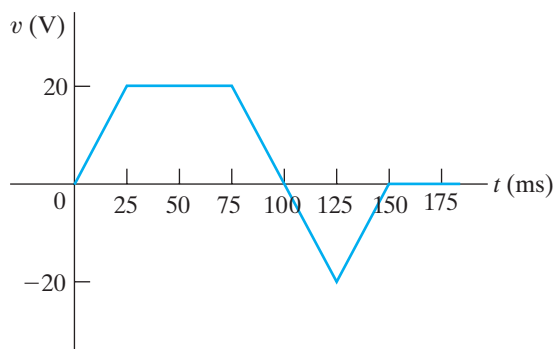
Figure P6.8



6.9 The current in and the voltage across a $10\ \text{H}$ inductor are known to be zero for $t \leq 0$. The voltage across the inductor is given by the graph in Fig. P6.9 for $t \geq 0$.

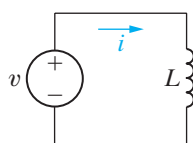
- a) Derive the expression for the current as a function of time in the intervals $0 \leq t \leq 25$ ms, $25 \text{ ms} \leq t \leq 75$ ms, $75 \text{ ms} \leq t \leq 125$ ms, $125 \text{ ms} \leq t \leq 150$ ms, and $150 \text{ ms} \leq t < \infty$.
- b) For $t > 0$, what is the current in the inductor when the voltage is zero?
- c) Sketch i versus t for $0 \leq t < \infty$.

Figure P6.9



- 6.10** a) Find the inductor current in the circuit in Fig. P6.10 if $v = 20 \cos 80t$ V, $L = 100$ mH, and $i(0) = 0$ A.
- b) Sketch v , i , p , and w versus t . In making these sketches, use the format used in Fig. 6.8. Plot over one complete cycle of the voltage waveform.
- c) Describe the subintervals in the time interval between 0 and 8π ms when power is being absorbed by the inductor. Repeat for the subintervals when power is being delivered by the inductor.

Figure P6.10

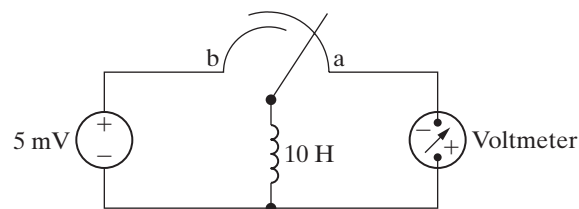


- 6.11** The current in a 100 mH inductor is known to be -10 A for $t \leq 0$ and $(-10 \cos 400t - 15 \sin 400t)e^{-200t}$ A for $t \geq 0$. Assume the passive sign convention.

- a) At what instant of time is the voltage across the inductor maximum?
- b) What is the maximum voltage?

- 6.12** In the circuit in Fig. P6.12, there was no energy stored initially in the 10 H inductor when it was placed across the terminals of the voltmeter. At $t = 0$ the inductor was switched instantaneously to position b where it remained for 1.6 s before returning instantaneously to position a. The d'Arsonval voltmeter has a full-scale reading of 20 V and a sensitivity of $1000 \Omega/\text{V}$. What will the reading of the voltmeter be at the instant the switch returns to position a if the inertia of d'Arsonval movement is negligible?

Figure P6.12



Section 6.2

- 6.13** The voltage across an $8 \mu\text{F}$ capacitor is known to be

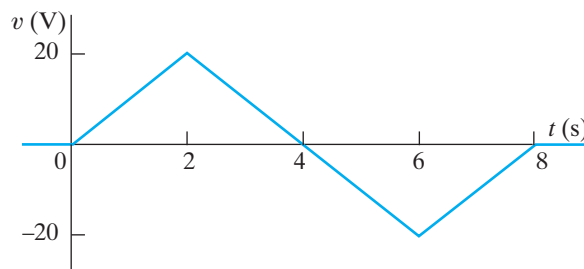
$$v_c = 600te^{-2500t} \text{ V for } t \geq 0.$$

- a) Find the current through the capacitor for $t > 0$. Assume the passive sign convention.
- b) Find the power at the terminals of the capacitor when $t = 100 \mu\text{s}$.
- c) Is the capacitor absorbing or delivering power at $t = 100 \mu\text{s}$.
- d) Find the energy stored in the capacitor at $t = 100 \mu\text{s}$.
- e) Find the maximum energy stored in the capacitors and the time when the maximum occurs.

- 6.14** The triangular voltage pulse shown in Fig. P6.14 is applied to a $200 \mu\text{F}$ capacitor.

- a) Write the expressions that describe $v(t)$ in the five time intervals $t < 0$, $0 \leq t \leq 2$ s, $2 \leq t \leq 6$ s, $6 \leq t \leq 8$ s, and $t > 8$ s.
- b) Derive the expressions for the capacitor current, power, and energy for the time intervals in part (a). Use the passive sign convention.
- c) Identify the time intervals between 0 and 8 s when power is being delivered by the capacitor. Repeat for the time intervals when power is being absorbed by the capacitor.

Figure P6.14



- 6.15** The voltage across the terminals of a $10 \mu\text{F}$ capacitor is

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$$v = \begin{cases} 70 \text{ V}, & t \leq 0; \\ (A_1 e^{-1400t} + A_2 t e^{-1400t}) \text{ V}, & t \geq 0. \end{cases}$$

The initial current in the capacitor is 150 mA. Assume the passive sign convention.

- What is the initial energy stored in the capacitor?
- Evaluate the coefficients A_1 and A_2 .
- What is the expression for the capacitor current?

6.16 A $100\ \mu\text{F}$ capacitor is subjected to a voltage pulse having a duration of 4 s. The pulse is described by the following equations:

$$v_c(t) = \begin{cases} 5t^3\ \text{V}, & 0 \leq t \leq 2\ \text{s}; \\ -5(t-4)^3\ \text{V}, & 2\ \text{s} \leq t \leq 4\ \text{s}; \\ 0 & \text{elsewhere.} \end{cases}$$

Sketch the current pulse that exists in the capacitor during the 4 s interval.

6.17 The voltage at the terminals of the capacitor in Fig. 6.10 is known to be

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$$v = \begin{cases} 50\ \text{V}, & t \leq 0; \\ 20 + 4e^{-300t}(5 \cos 1000t + \sin 1000t)\ \text{V}, & t \geq 0. \end{cases}$$

Assume $C = 125\ \mu\text{F}$.

- Find the current in the capacitor for $t < 0$.
- Find the current in the capacitor for $t > 0$.
- Is there any instantaneous change in the voltage across the capacitor at $t = 0$?
- Is there an instantaneous change in the current in the capacitor at $t = 0$?
- How much energy (in millijoules) is stored in the capacitor at $t = \infty$?

6.18 The expressions for voltage, power, and energy derived in Example 6.5 involved both integration and manipulation of algebraic expressions. As an engineer, you cannot accept such results on faith alone. That is, you should develop the habit of asking yourself, “Do these results make sense in terms of the known behavior of the circuit they purport to describe?” With these thoughts in mind, test the expressions of Example 6.5 by performing the following checks:

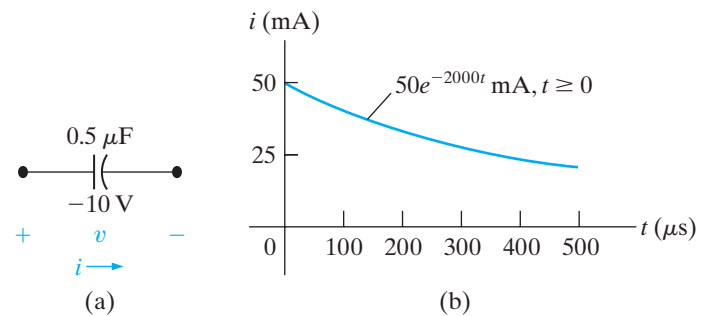
- Check the expressions to see whether the voltage is continuous in passing from one time interval to the next.
- Check the power expression in each interval by selecting a time within the interval and seeing whether it gives the same result as the corresponding product of v and i . For example, test at 10 and 30 μs .

- Check the energy expression within each interval by selecting a time within the interval and seeing whether the energy equation gives the same result as $\frac{1}{2} C v^2$. Use 10 and 30 μs as test points.

6.19 The initial voltage on the $0.5\ \mu\text{F}$ capacitor shown in Fig. P6.19(a) is $-10\ \text{V}$. The capacitor current has the waveform shown in Fig. P6.19(b).

- How much energy, in microjoules, is stored in the capacitor at $t = 200\ \mu\text{s}$?
- Repeat (a) for $t = \infty$.

Figure P6.19

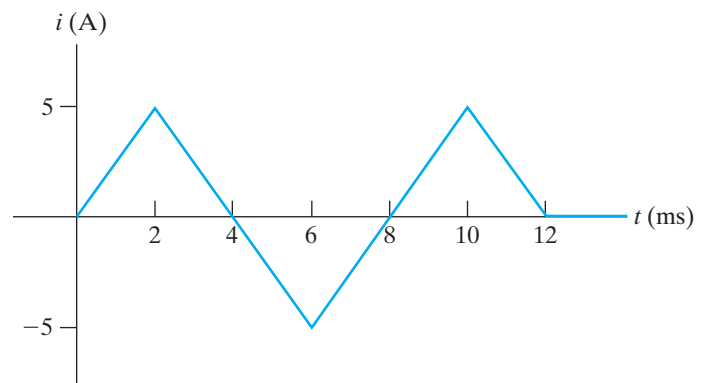


6.20 The current shown in Fig. P6.20 is applied to a $3\ \mu\text{F}$ capacitor. The initial voltage on the capacitor is zero.

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- Find the charge on the capacitor at $t = 6\ \text{ms}$.
- Find the voltage on the capacitor at $t = 10\ \text{ms}$.
- How much energy is stored in the capacitor by this current?

Figure P6.20

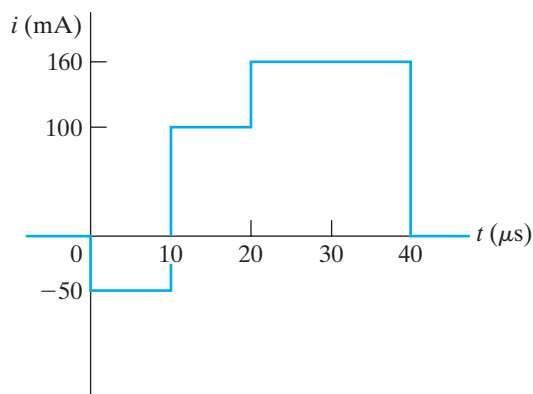


6.21 The rectangular-shaped current pulse shown in Fig. P6.21 is applied to a $0.1\ \mu\text{F}$ capacitor. The initial voltage on the capacitor is a 15 V drop in the reference direction of the current. Derive the expression for the capacitor voltage for the time intervals in (a)–(d).

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- $0 \leq t \leq 10 \mu\text{s}$;
- $10 \mu\text{s} \leq t \leq 20 \mu\text{s}$;
- $20 \mu\text{s} \leq t \leq 40 \mu\text{s}$
- $40 \mu\text{s} \leq t < \infty$
- Sketch $v(t)$ over the interval $-10 \mu\text{s} \leq t \leq 50 \mu\text{s}$.

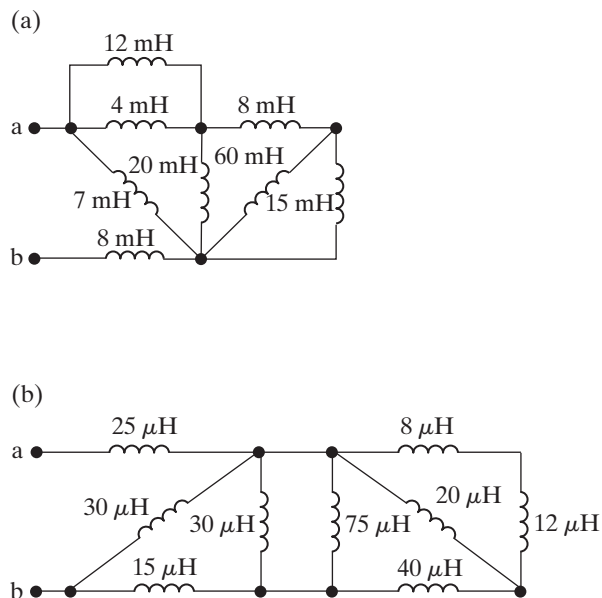
Figure P6.21



Section 6.3

- 6.22** Assume that the initial energy stored in the inductors of Figs. P6.22(a) and (b) is zero. Find the equivalent inductance with respect to the terminals a, b.

Figure P6.22



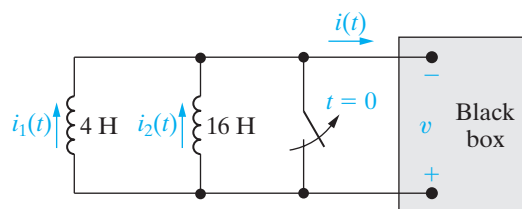
- 6.23** Use realistic inductor values from Appendix H to construct series and parallel combinations of inductors to yield the equivalent inductances specified below. Try to minimize the number of inductors used. Assume that no initial energy is stored in any of the inductors.

- 8 mH;
- $45 \mu\text{H}$;
- $180 \mu\text{H}$.

- 6.24** The two parallel inductors in Fig. P6.24 are connected across the terminals of a black box at $t = 0$. The resulting voltage v for $t > 0$ is known to be $64e^{-4t}$ V. It is also known that $i_1(0) = -10$ A and $i_2(0) = 5$ A.

- Replace the original inductors with an equivalent inductor and find $i(t)$ for $t \geq 0$.
- Find $i_1(t)$ for $t \geq 0$.
- Find $i_2(t)$ for $t \geq 0$.
- How much energy is delivered to the black box in the time interval $0 \leq t < \infty$?
- How much energy was initially stored in the parallel inductors?
- How much energy is trapped in the ideal inductors?
- Show that your solutions for i_1 and i_2 agree with the answer obtained in (f).

Figure P6.24



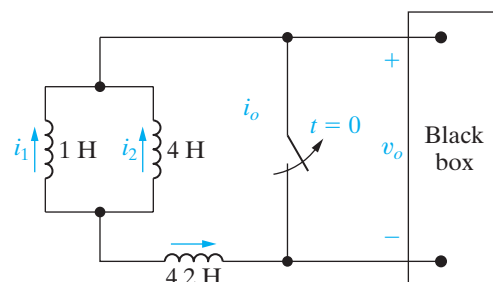
- 6.25** The three inductors in the circuit in Fig. P6.25 are connected across the terminals of a black box at $t = 0$. The resulting voltage for $t > 0$ is known to be

$$v_o = 500e^{-100t} \text{ V.}$$

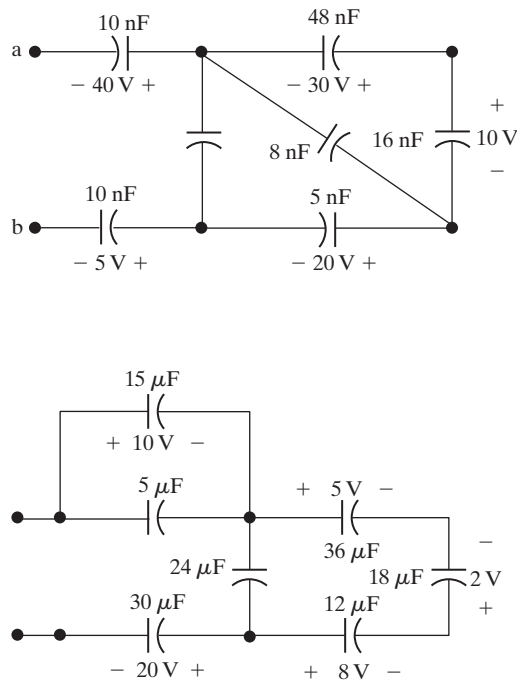
If $i_1(0) = -6$ A and $i_2(0) = 1$ A, find

- $i_o(0)$;
- $i_o(t)$, $t \geq 0$;
- $i_1(t)$, $t \geq 0$;
- $i_2(t)$, $t \geq 0$;
- the initial energy stored in the three inductors;
- the total energy delivered to the black box; and
- the energy trapped in the ideal inductors.

Figure P6.25



- 6.26** For the circuit shown in Fig. P6.25, how many milliseconds after the switch is opened is the energy delivered to the black box 80% of the total energy delivered?
- 6.27** Find the equivalent capacitance with respect to the terminals a, b for the circuits shown in Fig. P6.27.

Figure P6.27

- 6.28** Use realistic capacitor values from Appendix H to construct series and parallel combinations of capacitors to yield the equivalent capacitances specified below. Try to minimize the number of capacitors used. Assume that no initial energy is stored in any of the capacitors.

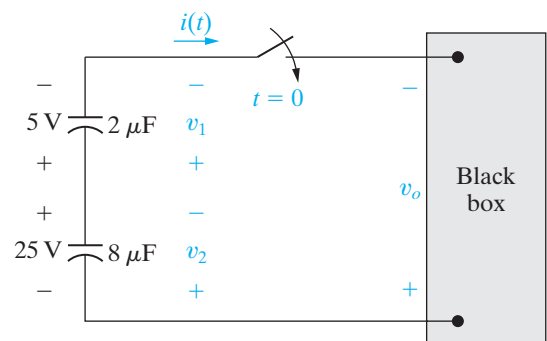
- 480 pF;
- 600 nF;
- 120 μ F.

- 6.29** Derive the equivalent circuit for a series connection of ideal capacitors. Assume that each capacitor has its own initial voltage. Denote these initial voltages as $v_1(t_0)$, $v_2(t_0)$, and so on. (Hint: Sum the voltages across the string of capacitors, recognizing that the series connection forces the current in each capacitor to be the same.)
- 6.30** Derive the equivalent circuit for a parallel connection of ideal capacitors. Assume that the initial voltage across the paralleled capacitors is $v(t_0)$. (Hint: Sum the currents into the string of capacitors, recognizing that the parallel connection forces the voltage across each capacitor to be the same.)

- 6.31** The two series-connected capacitors in Fig. P6.31 are connected to the terminals of a black box at

$t = 0$. The resulting current $i(t)$ for $t > 0$ is known to be $800e^{-25t} \mu\text{A}$.

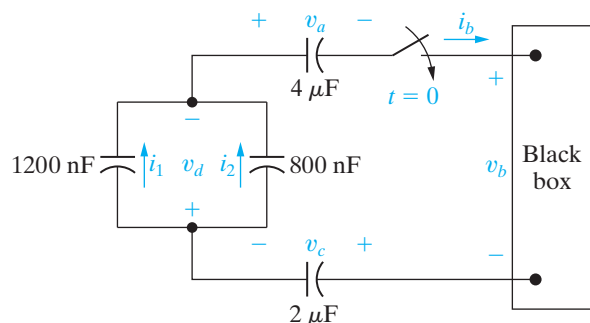
- Replace the original capacitors with an equivalent capacitor and find $v_o(t)$ for $t \geq 0$.
- Find $v_1(t)$ for $t \geq 0$.
- Find $v_2(t)$ for $t \geq 0$.
- How much energy is delivered to the black box in the time interval $0 \leq t < \infty$?
- How much energy was initially stored in the series capacitors?
- How much energy is trapped in the ideal capacitors?
- Show that the solutions for v_1 and v_2 agree with the answer obtained in (f).

Figure P6.31

- 6.32** The four capacitors in the circuit in Fig. P6.32 are connected across the terminals of a black box at $t = 0$. The resulting current i_b for $t > 0$ is known to be

$$i_b = -5e^{-50t} \text{ mA}.$$

If $v_a(0) = -20 \text{ V}$, $v_c(0) = -30 \text{ V}$ and $v_d(0) = 250 \text{ V}$, find the following for $t \geq 0$: (a) $v_b(t)$, (b) $v_a(t)$, (c) $v_c(t)$, (d) $v_d(t)$, (e) $i_1(t)$, and (f) $i_2(t)$.

Figure P6.32

- 6.33** For the circuit in Fig. P6.32, calculate
- the initial energy stored in the capacitors;
 - the final energy stored in the capacitors;

Problems

Section 9.1

9.1 A sinusoidal current is given by the expression

$$i = 100 \cos(600t + 45^\circ) \text{ mA.}$$

Find (a) f in hertz; (b) T in milliseconds; (c) I_m ; (d) $i(0)$; (e) ϕ in degrees and radians; (f) the smallest positive value of t at which $i = 0$; and (g) the smallest positive value of t at which $di/dt = 0$.

9.2 In a single graph, sketch $v = 100 \cos(\omega t + \phi)$ versus ωt for $\phi = -60^\circ, -30^\circ, 0^\circ, 30^\circ$, and 60° .

- State whether the voltage function is shifting to the right or left as ϕ becomes more positive.
- What is the direction of shift if ϕ changes from 0 to -30° ?

9.3 Consider the sinusoidal voltage

$$v(t) = 25 \cos(400\pi t + 60^\circ) \text{ V.}$$

- What is the maximum amplitude of the voltage?
- What is the frequency in hertz?
- What is the frequency in radians per second?
- What is the phase angle in radians?
- What is the phase angle in degrees?
- What is the period in milliseconds?
- What is the first time after $t = 0$ that $v = 0$ V?
- The sinusoidal function is shifted $5/6$ ms to the right along the time axis. What is the expression for $v(t)$?
- What is the minimum number of milliseconds that the function must be shifted to the left if the expression for $v(t)$ is $25 \sin 400\pi t$ V?

9.4 A series circuit with $R = 20 \Omega$ and $C = 200 \text{ pF}$ has a sinusoidal voltage with a frequency of 99.47 MHz applied to it. If the maximum voltage across the capacitor is 24 V , what is the maximum voltage across the series combination?

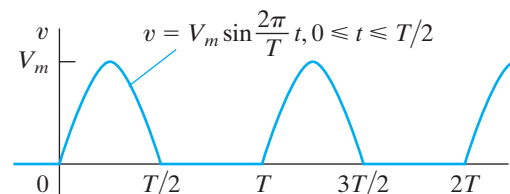
9.5 At $t = 5 \text{ ms}$, a sinusoidal current is known to be zero and going negative. The current is next zero at $t = 25 \text{ ms}$. It is also known that the current is 50 mA at $t = 0$.

- What is the frequency of i in hertz?
- What is the expression for i ?

9.6 The rms value of the sinusoidal voltage supplied to the convenience outlet of a home in Scotland is 240 V . What is the maximum value of the voltage at the outlet?

9.7 Find the rms value of the half-wave rectified sinusoidal voltage shown in Fig. P9.7.

Figure P9.7



9.8 Show that

$$\int_{t_0}^{t_0+T} V_m^2 \cos^2(\omega t + \phi) dt = \frac{V_m^2 T}{2}.$$

Section 9.2

9.9 The voltage applied to the circuit shown in Fig. 9.5 at $t = 0$ is $75 \cos(4000t - 60^\circ) \text{ V}$. The circuit resistance is 400Ω and the initial current in the 75 mH inductor is zero.

- Find $i(t)$ for $t \geq 0$.
- Write the expressions for the transient and steady-state components of $i(t)$.
- Find the numerical value of i after the switch has been closed for $750 \mu\text{s}$.
- What are the maximum amplitude, frequency (in radians per second), and phase angle of the steady-state current?
- By how many degrees are the voltage and the steady-state current out of phase?

9.10 a) Verify that Eq. 9.7 is the solution of Eq. 9.6. This can be done by substituting Eq. 9.7 into the left-hand side of Eq. 9.6 and then noting that it equals the right-hand side for all values of $t > 0$. At $t = 0$, Eq. 9.7 should reduce to the initial value of the current.

- Because the transient component vanishes as time elapses and because our solution must satisfy the differential equation for all values of t , the steady-state component, by itself, must also satisfy the differential equation. Verify this observation by showing that the steady-state component of Eq. 9.7 satisfies Eq. 9.6.

Sections 9.3–9.4

9.11 Use the concept of the phasor to combine the following sinusoidal functions into a single trigonometric expression:

- $y = 30 \cos(200t - 160^\circ) + 15 \cos(200t + 70^\circ)$,
- $y = 90 \sin(50t - 20^\circ) + 60 \cos(50t - 70^\circ)$,

- c) $y = 50 \cos(5000t - 60^\circ) + 25 \sin(5000t + 110^\circ) - 75 \cos(5000t - 30^\circ)$,
 d) $y = 10 \cos(\omega t + 30^\circ) + 10 \sin \omega t + 10 \cos(\omega t + 150^\circ)$.

9.12 A 200 Hz sinusoidal voltage with a maximum amplitude of 100 V at $t = 0$ is applied across the terminals of an inductor of 2 mH. Find the maximum amplitude of the steady-state current in the inductor.

- What is the frequency of the inductor current?
- If the phase angle of the voltage is zero, what is the phase angle of the current?
- What is the inductive reactance of the inductor?
- Calculate the current through the inductor in amperes.
- What is the impedance of the inductor?

9.13 A 80 kHz sinusoidal voltage has zero phase angle and a maximum amplitude of 25 mV. When this voltage is applied across the terminals of a capacitor, the resulting steady-state current has a maximum amplitude of 628.32 μ A.

- What is the frequency of the current in radians per second?
- What is the phase angle of the current?
- What is the capacitive reactance of the capacitor?
- What is the capacitance of the capacitor in microfarads?
- What is the impedance of the capacitor?

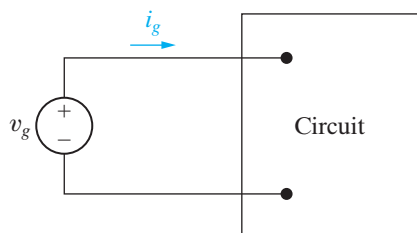
9.14 The expressions for the steady-state voltage and current at the terminals of the circuit seen in Fig. P9.14 are

$$v_g = 100 \cos(1000\pi t) \text{ V},$$

$$i_g = 2 \sin(1000\pi t + 40^\circ) \text{ A}.$$

- What is the impedance seen by the source?
- By how many microseconds is the current out of phase with the voltage?

Figure P9.14



Sections 9.5 and 9.6

9.15 A 25 Ω resistor, a 50 mH inductor, and a 32 μ F capacitor are connected in series. The series-connected elements are energized by a sinusoidal voltage source whose voltage is $25 \cos(500t - 60^\circ)$ V.

- Draw the frequency-domain equivalent circuit.
- Reference the current in the direction of the voltage rise across the source, and find the phasor current.
- Find the steady-state expression for $i(t)$.

9.16 A 20 Ω resistor and a 0.5 mH inductor are connected in parallel. This parallel combination is also in parallel with the series combination of a 16 Ω resistor and a 600 μ H inductor. These three parallel branches are driven by a sinusoidal current source whose current is $922 \cos(20,000t + 30^\circ)$ A.

- Draw the frequency-domain equivalent circuit.
- Reference the voltage across the current source as a rise in the direction of the source current, and find the phasor voltage.
- Find the steady-state expression for $v(t)$.

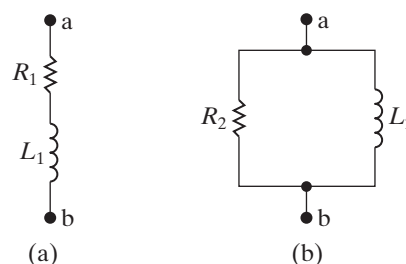
9.17 Three branches having impedances of $3 + j4 \Omega$, $16 - j12 \Omega$, and $-j4 \Omega$, respectively, are connected in parallel. What are the equivalent (a) admittance, (b) conductance, and (c) susceptance of the parallel connection in millisiemens? (d) If the parallel branches are excited from a sinusoidal current source where $i = 8 \cos \omega t$ A, what is the maximum amplitude of the current in the purely capacitive branch?

9.18 a) Show that, at a given frequency ω , the circuits in Fig. P9.18(a) and (b) will have the same impedance between the terminals a,b if

$$R_1 = \frac{\omega^2 L_2^2 R_2}{R_2^2 + \omega^2 L_2^2}, \quad L_1 = \frac{R_2^2 L_2}{R_2^2 + \omega^2 L_2^2}.$$

- b) Find the values of resistance and inductance that when connected in series will have the same impedance at 4 krad/s as that of a 5 k Ω resistor connected in parallel with a 1.25 H inductor.

Figure P9.18



9.19 a) Show that at a given frequency ω , the circuits in Fig. P9.18(a) and (b) will have the same impedance between the terminals a,b if

$$R_2 = \frac{R_1^2 + \omega^2 L_1^2}{R_1}, \quad L_2 = \frac{R_1^2 + \omega^2 L_1^2}{\omega^2 L_1}.$$

(Hint: The two circuits will have the same impedance if they have the same admittance.)

- b) Find the values of resistance and inductance that when connected in parallel will have the same impedance at 1 krad/s as an 8 k Ω resistor connected in series with a 4 H inductor.

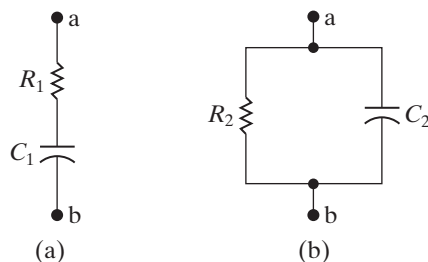
9.20 a) Show that at a given frequency ω , the circuits in Fig. P9.20(a) and (b) will have the same impedance between the terminals a,b if

$$R_1 = \frac{R_2}{1 + \omega^2 R_2^2 C_2^2},$$

$$C_1 = \frac{1 + \omega^2 R_2^2 C_2^2}{\omega^2 R_2^2 C_2}.$$

- b) Find the values of resistance and capacitance that when connected in series will have the same impedance at 40 krad/s as that of a 1000 Ω resistor connected in parallel with a 50 nF capacitor.

Figure P9.20



9.21 a) Show that at a given frequency ω , the circuits in Fig. 9.20(a) and (b) will have the same impedance between the terminals a,b if

$$R_2 = \frac{1 + \omega^2 R_1^2 C_1^2}{\omega^2 R_1 C_1^2},$$

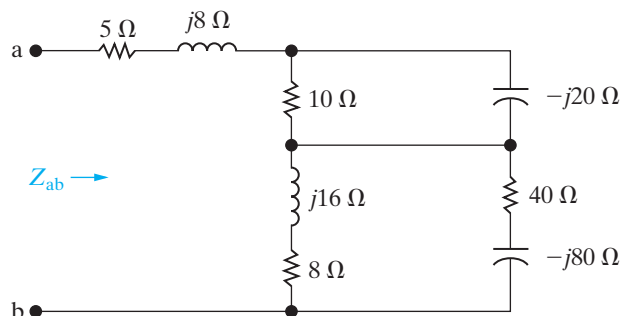
$$C_2 = \frac{C_1}{1 + \omega^2 R_1^2 C_1^2}.$$

(Hint: The two circuits will have the same impedance if they have the same admittance.)

- b) Find the values of resistance and capacitance that when connected in parallel will give the same impedance at 50 krad/s as that of a 1 k Ω resistor connected in series with a capacitance of 40 nF.

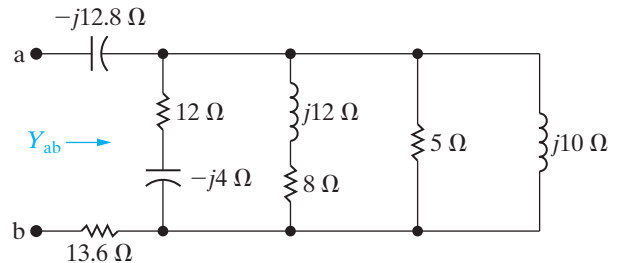
9.22 Find the impedance Z_{ab} in the circuit seen in Fig. P9.22. Express Z_{ab} in both polar and rectangular form.

Figure P9.22



9.23 Find the admittance Y_{ab} in the circuit seen in Fig. P9.23. Express Y_{ab} in both polar and rectangular form. Give the value of Y_{ab} in millisiemens.

Figure P9.23

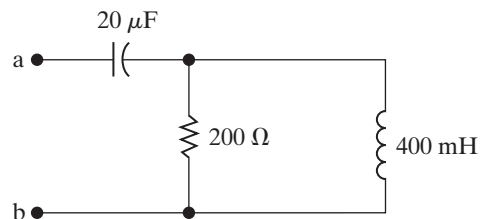


9.24 a) For the circuit shown in Fig. P9.24, find the frequency (in radians per second) at which the impedance Z_{ab} is purely resistive.

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- b) Find the value of Z_{ab} at the frequency of (a).

Figure P9.24



9.25 a) Using component values from Appendix H, combine at least one resistor, inductor, and capacitor in series to create an impedance of $300 - j400 \Omega$ at a frequency of 10,000 rad/s.

- b) At what frequency does the circuit from part (a) have an impedance that is purely resistive?

9.26 a) Using component values from Appendix H, combine at least one resistor and one inductor in parallel to create an impedance of $40 + j20 \Omega$ at a frequency of 5000 rad/s. (Hint: Use the results of Problem 9.19.)

- b) Using component values from Appendix H, combine at least one resistor and one capacitor in parallel to create an impedance of $40 - j20 \Omega$ at a frequency of 5000 rad/s. (Hint: Use the result of Problem 9.21.)

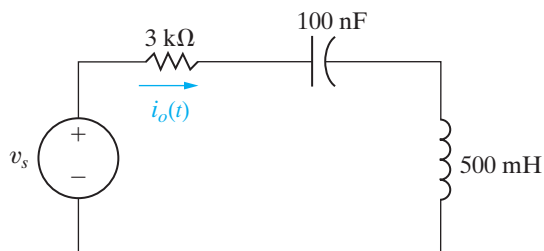
9.27 a) Using component values from Appendix H, find a single capacitor or a network of capacitors that, when combined in parallel with the RL circuit from Problem 9.26(a), gives an equivalent impedance that is purely resistive at a frequency of 5000 rad/s.

- b) Using component values from Appendix H, find a single inductor or a network of inductors that, when combined in parallel with the RC circuit from Problem 9.26(b), gives an equivalent impedance that is purely resistive at a frequency of 5000 rad/s.

- 9.28** Find the steady-state expression for $i_o(t)$ in the circuit in Fig. P9.28 if $v_s = 80 \cos 2000t$ V.

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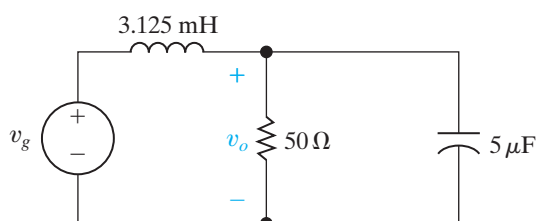
Figure P9.28



- 9.29** The circuit in Fig. P9.29 is operating in the sinusoidal steady state. Find the steady-state expression for $v_o(t)$ if $v_g = 60 \sin 8000t$ V.

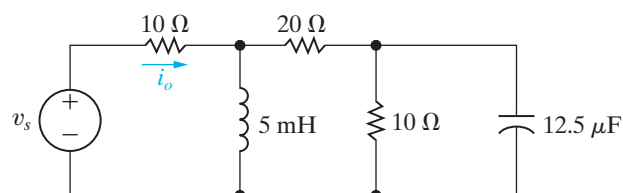
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Figure P9.29



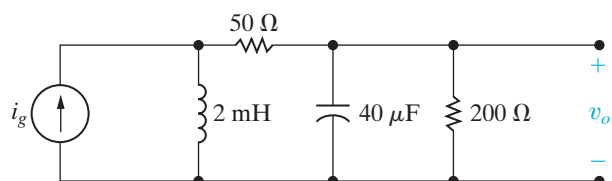
- 9.30** The circuit in Fig. P9.30 is operating in the sinusoidal steady state. Find $i_o(t)$ if $v_s(t) = 25 \sin 2000t$ V.

Figure P9.30



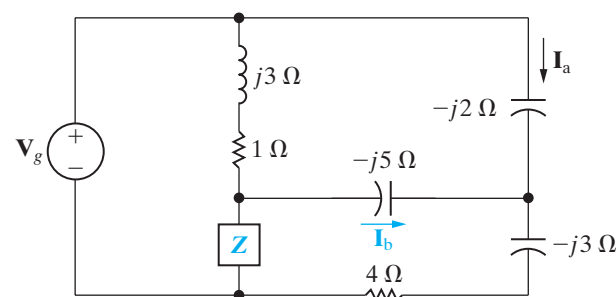
- 9.31** a) For the circuit shown in Fig. P9.31, find the steady-state expression for v_o if $i_g = 25 \cos 50,000t$ mA.
b) By how many microseconds does v_o lead i_g ?

Figure P9.31



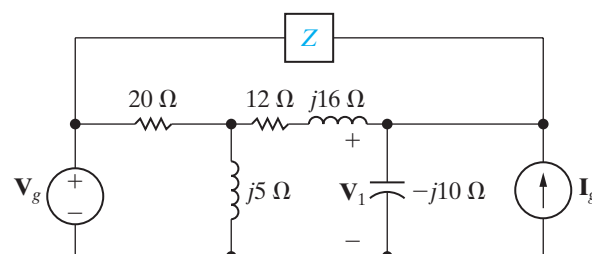
- 9.32** Find $\mathbf{I}_b$ and $\mathbf{Z}$ in the circuit shown in Fig. P9.32 if $\mathbf{V}_g = 25 \angle 0^\circ$ V and $\mathbf{I}_a = 5 \angle 90^\circ$ A.

Figure P9.32



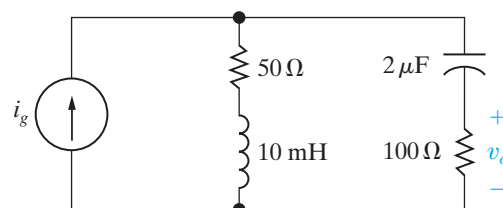
- 9.33** Find the value of $\mathbf{Z}$ in the circuit seen in Fig. P9.33 if $\mathbf{V}_g = 100 - j50$ V, $\mathbf{I}_g = 30 + j20$ A, and $\mathbf{V}_1 = 140 + j30$ V.

Figure P9.33



- 9.34** Find the steady-state expression for v_o in the circuit of Fig. P9.34 if $i_g = 60 \cos 10,000t$ mA.

Figure P9.34

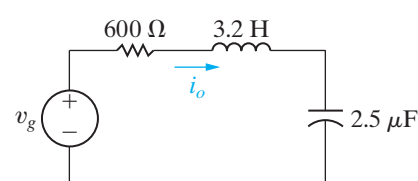


- 9.35** The circuit shown in Fig. P9.35 is operating in the sinusoidal steady state. Find the value of ω if

$$i_o = 40 \sin(\omega t + 21.87^\circ) \text{ mA},$$

$$v_g = 40 \cos(\omega t - 15^\circ) \text{ V}.$$

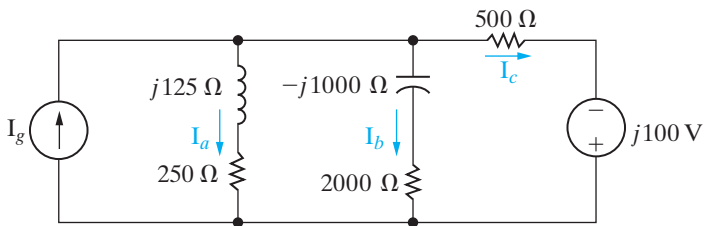
Figure P9.35



- 9.36** The phasor current $\mathbf{I}_b$ in the circuit shown in Fig. P9.36 is $25 \angle 0^\circ$ mA.

- a) Find $\mathbf{I}_a$, $\mathbf{I}_c$, and $\mathbf{I}_g$.
 b) If $\omega = 2000$ rad/s, write expressions for $i_a(t)$, $i_c(t)$, and $i_g(t)$.

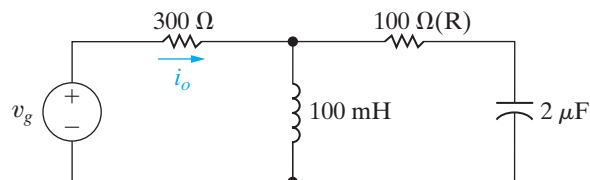
Figure P9.36



9.37 The frequency of the sinusoidal voltage source in the circuit in Fig. P9.37 is adjusted until the current i_o is in phase with v_g .

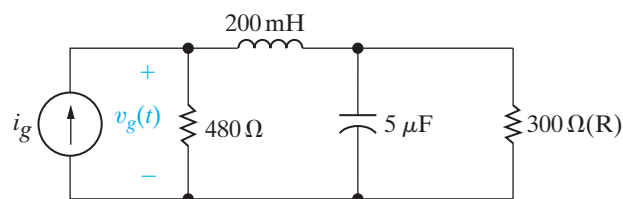
- a) Find the frequency in hertz.
 b) Find the steady-state expression for i_g (at the frequency found in [a]) if $v_g = 45 \cos \omega t$ V.
 c) Find the maximum value of 'R' for which resonant frequency of the circuit will have real value.

Figure P9.37



- 9.38** a) The frequency of the source voltage in the circuit in Fig. P9.38 is adjusted until v_g is in phase with i_g . What is the value of ω in radians per second?
 b) If $i_g = 80 \cos \omega t$ mA (where ω is the frequency found in [a]), what is the steady-state expression for v_g ?
 c) Find the minimum value of 'R' for which resonant frequency of the circuit will have real value.

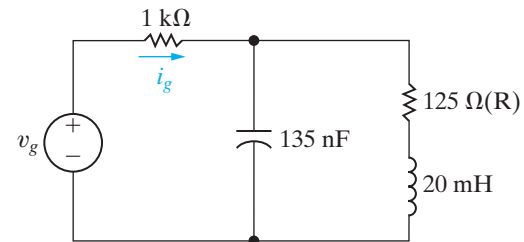
Figure P9.38



9.39 The frequency of the sinusoidal voltage source in the circuit in Fig. P9.39 is adjusted until i_g is in phase with v_g .

- a) What is the value of ω in radians per second?
 b) If $v_g = 15 \cos \omega t$ V (where ω is the frequency found in [a]), what is the steady-state expression for i_g ?
 c) Find the minimum value of 'R' for which resonant frequency of the circuit will have real value.

Figure P9.39

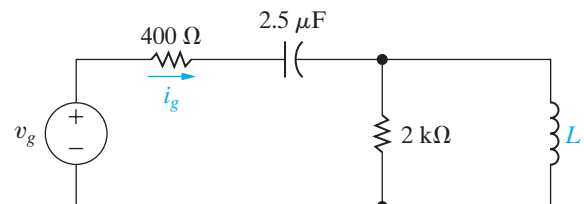


9.40 a) The source voltage in the circuit in Fig. P9.40 is $v_g = 200 \cos 500t$ V. Find the values of L such that i_g is in phase with v_g when the circuit is operating in the steady state.

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b) For the values of L found in (a), find the steady-state expressions for i_g .

Figure P9.40

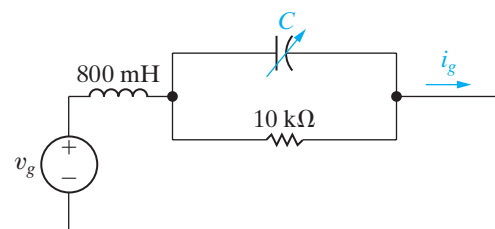


9.41 The circuit shown in Fig. P9.41 is operating in the sinusoidal steady state. The capacitor is adjusted until the current i_g is in phase with the sinusoidal voltage v_g .

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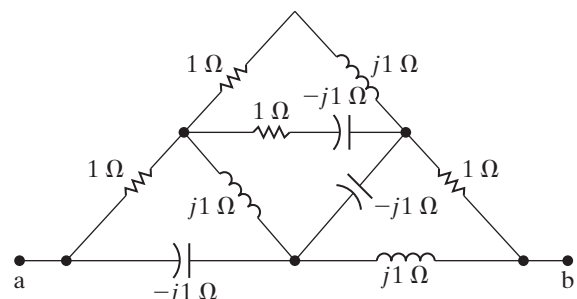
- a) Specify the capacitance in microfarads if $v_g = 80 \cos 5000t$ V.
 b) Give the steady-state expression for i_g when C has the value found in (a).

Figure P9.41



9.42 Find Z_{ab} for the circuit shown in Fig P9.42.

Figure P9.42

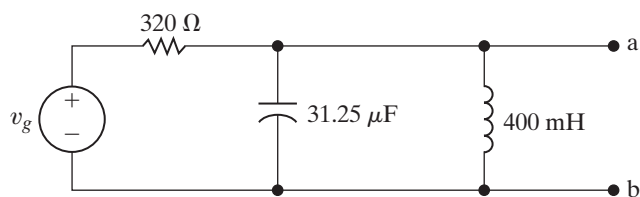


Section 9.7

9.43 The sinusoidal voltage source in the circuit in Fig. P9.43 is developing a voltage equal to $50 \sin 400t$ V.

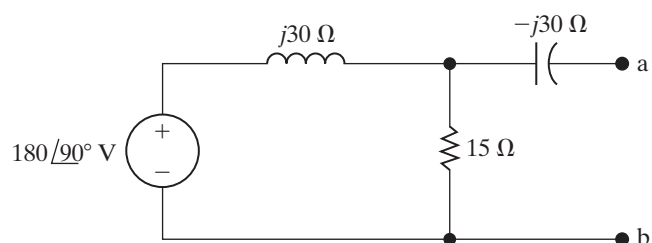
- Find the Thévenin voltage with respect to the terminals a,b.
- Find the Thévenin impedance with respect to the terminals a,b.
- Draw the Thévenin equivalent.

Figure P9.43



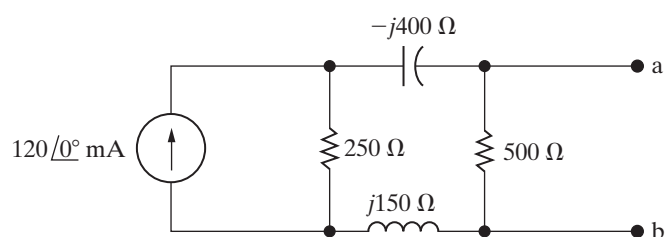
9.44 Use source transformations to find the Norton equivalent circuit with respect to the terminals a,b for the circuit shown in Fig. P9.44.

Figure P9.44



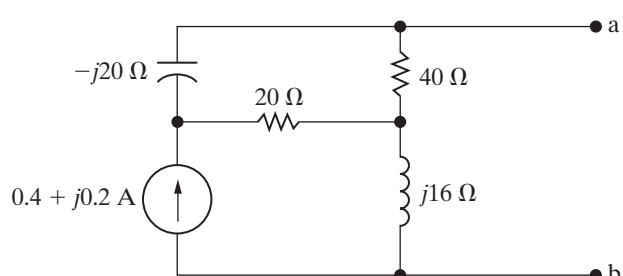
9.45 Use source transformations to find the Thévenin equivalent circuit with respect to the terminals a,b for the circuit shown in Fig. P9.45.

Figure P9.45



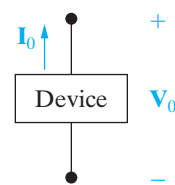
9.46 Find the Norton equivalent circuit with respect to the terminals a,b for the circuit shown in Fig. P9.46.

Figure P9.46



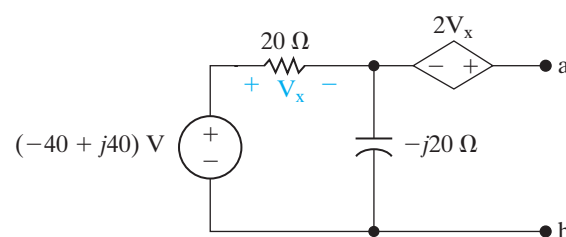
9.47 The device in Fig. P9.47 is represented in the frequency domain by a Norton equivalent. When a resistor having an impedance of $5 \text{ k}\Omega$ is connected across the device, the value of $\mathbf{V}_0$ is $(5 - j15) \text{ V}$. When a capacitor having an impedance of $-j3 \text{ k}\Omega$ is connected across the device, the value of $\mathbf{I}_0$ is $(4.5 - j6) \text{ mA}$. Find the Norton current $\mathbf{I}_N$ and the Norton impedance $\mathbf{Z}_N$.

Figure P9.47



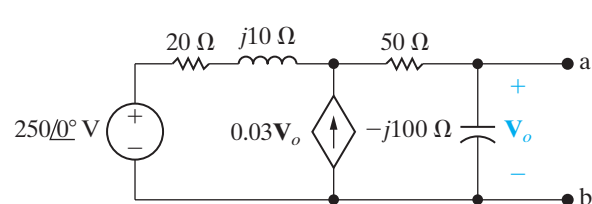
9.48 Find the Norton equivalent with respect to terminals a, b in the circuit of Fig. P9.48.

Figure P9.48



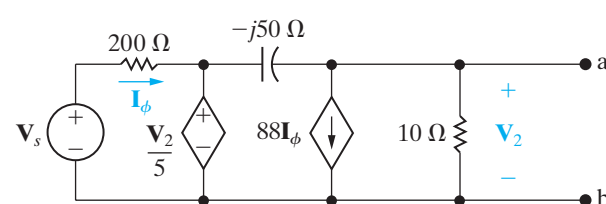
9.49 Find the Thévenin equivalent circuit with respect to the terminals a,b of the circuit shown in Fig. P9.49.

Figure P9.49



9.50 Find the Norton equivalent circuit with respect to the terminals a,b for the circuit shown in Fig. P9.50 when $\mathbf{V}_s = 5\angle 0^\circ \text{ V}$.

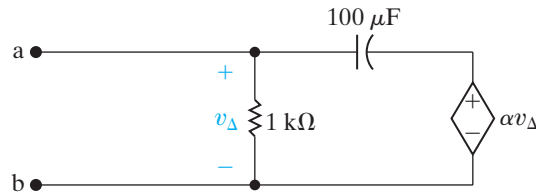
Figure P9.50



9.51 The circuit shown in Fig. P9.51 is operating at a frequency of 10 rad/s . Assume α is real and lies between -10 and $+10$, that is, $-10 \leq \alpha \leq 10$.

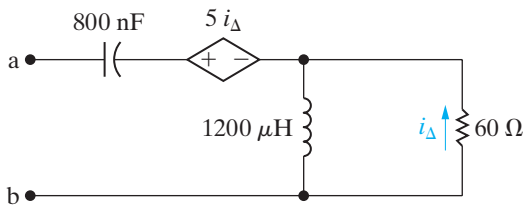
- Find the value of α so that the Thévenin impedance looking into the terminals a,b is purely resistive.
- What is the value of the Thévenin impedance for the α found in (a)?
- Can α be adjusted so that the Thévenin impedance equals $500 - j500 \Omega$? If so, what is the value of α ?
- For what values of α will the Thévenin impedance be inductive?

Figure P9.51



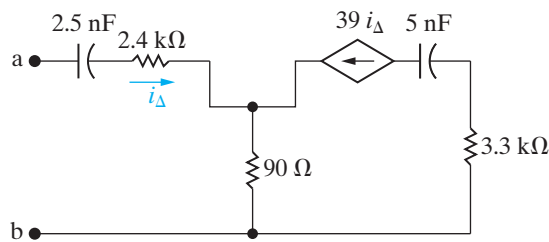
- 9.52** Find Z_{ab} in the circuit shown in Fig. P9.52 when the circuit is operating at a frequency of 100 krad/s.

Figure P9.52



- 9.53** Find the Thévenin impedance seen looking into the terminals a,b of the circuit in Fig. P9.53 if the frequency of operation is $(25/\pi)$ kHz.

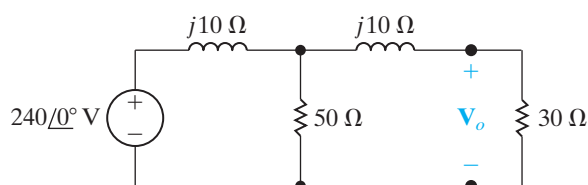
Figure P9.53



Section 9.8

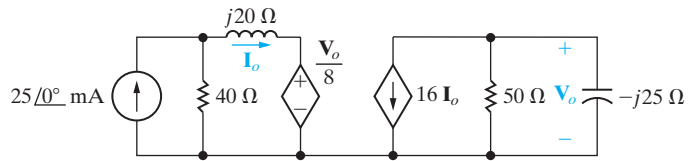
- 9.54** Use the node-voltage method to find V_o in the circuit in Fig. P9.54.

Figure P9.54



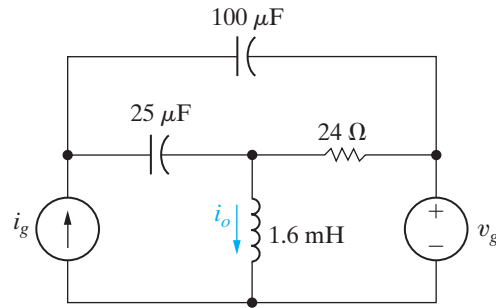
- 9.55** Use the node-voltage method to find V_o and I_o in the circuit seen in Fig. P9.55.

Figure P9.55



- 9.56** Use the node-voltage method to find the steady-state expression for i_o in the circuit seen in Fig. P9.56 if $i_g = 5 \cos 2500t$ A and $v_g = 20 \cos (2500t + 90^\circ)$ V.

Figure P9.56

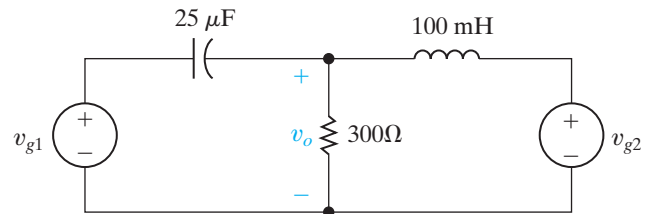


- 9.57** Use the node-voltage method to find the steady-state expression for $v_o(t)$ in the circuit in Fig. P9.57 if

$$v_{g1} = 20 \sin (400t + 143.13^\circ) \text{ V},$$

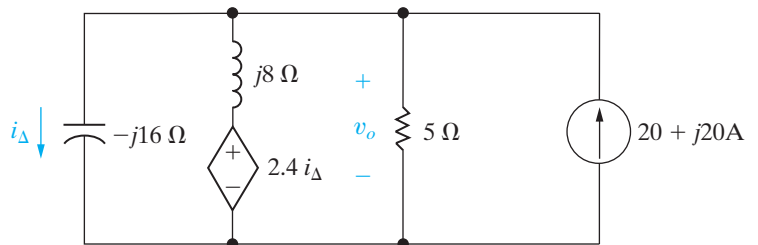
$$v_{g2} = 6.01 \cos (400t + 33.69^\circ) \text{ V}.$$

Figure P9.57



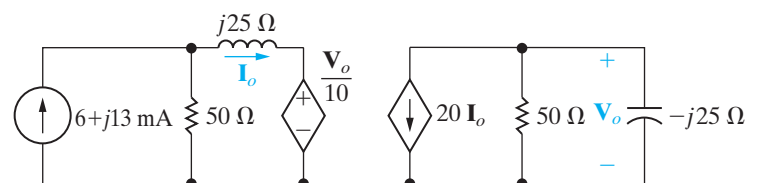
- 9.58** Use the node-voltage method to find the phasor voltage V_o in the circuit shown in Fig. P9.58. Express the voltage in both polar and rectangular form.

Figure P9.58



- 9.59** Use the node-voltage method to find V_o and I_o in the circuit seen in Fig. P9.59.

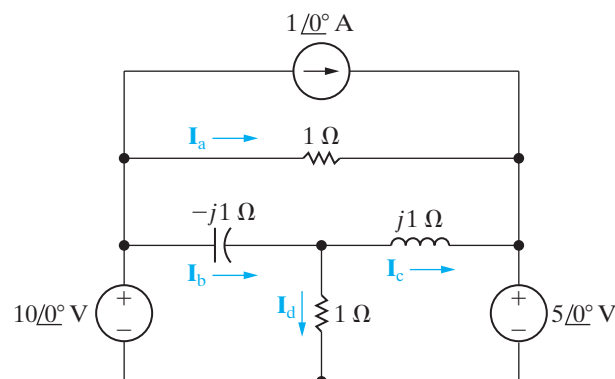
Figure P9.59



Section 9.9

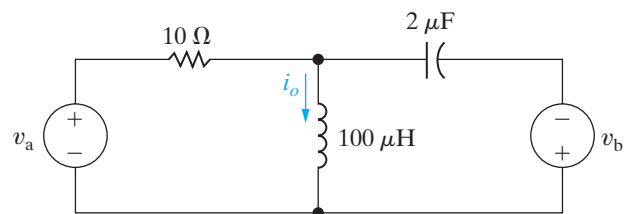
- 9.60** Use the mesh-current method to find the phasor current $\mathbf{I}_g$ in the circuit in Fig. P9.55.
- 9.61** Use the mesh-current method to find the steady-state expression for $i_o(t)$ in the circuit in Fig. P9.56.
- 9.62** Use the mesh-current method to find the branch currents $\mathbf{I}_a$, $\mathbf{I}_b$, $\mathbf{I}_c$, and $\mathbf{I}_d$ in the circuit shown in Fig. P9.62.

Figure P9.62



- 9.63** Use the mesh-current method to find the steady-state expression for $i_o(t)$ in the circuit in Fig. P9.63 if
- $$v_a = 100 \cos 50,000t \text{ V},$$
- $$v_b = 100 \sin(50,000t + 180^\circ) \text{ V}.$$

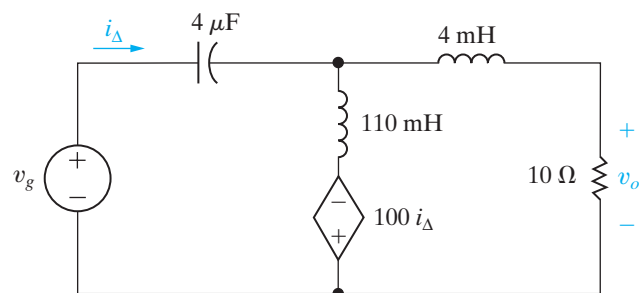
Figure P9.63



- 9.64** Use the mesh-current method to find the steady-state expression for v_o in the circuit seen in Fig. P9.64 if v_g equals $75 \cos 5000t \text{ V}$.

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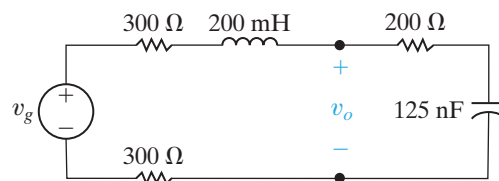
Figure P9.64



Sections 9.5–9.9

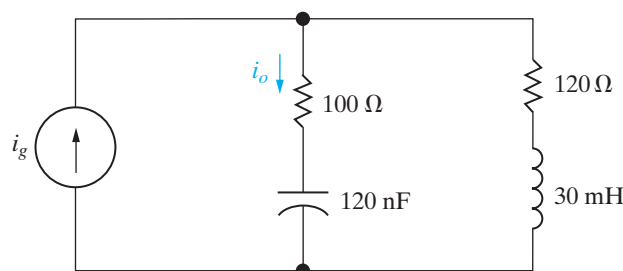
- 9.65** Use voltage division to find the steady-state expression for $v_o(t)$ in the circuit in Fig. P9.65 if $v_g = 100 \cos 8000t \text{ V}$.

Figure P9.65



- 9.66** Use current division to find the steady-state expression for i_o in the circuit in Fig. P9.66 if $i_g = 400 \cos 20,000t \text{ mA}$.

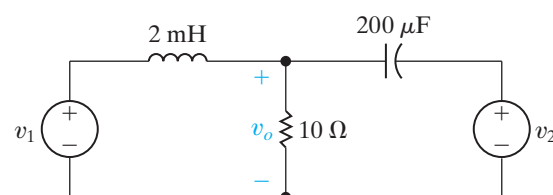
Figure P9.66



- 9.67** For the circuit in Fig. P9.67, suppose
- $$v_1 = 20 \cos(2000t - 36.87^\circ) \text{ V},$$
- $$v_2 = 10 \cos(5000t + 16.26^\circ) \text{ V}.$$

- a) What circuit analysis technique must be used to find the steady-state expression for $v_o(t)$?
- b) Find the steady-state expression for $v_o(t)$.

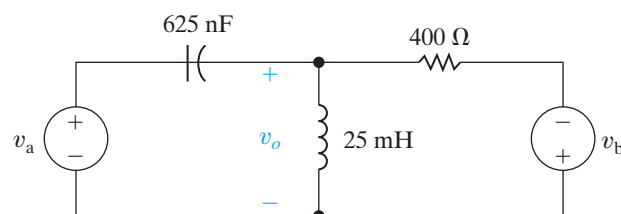
Figure P9.67



- 9.68** For the circuit in Fig. P9.68, suppose
- $$v_a = 10 \cos 16,000t \text{ V},$$
- $$v_b = 20 \cos 4000t \text{ V}.$$

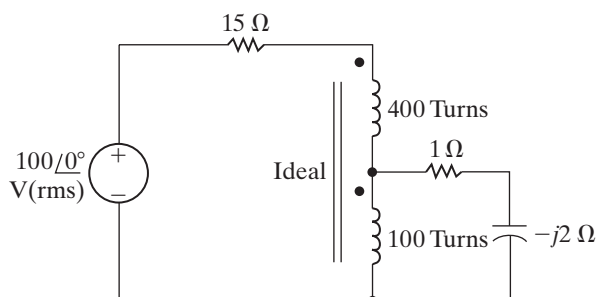
- a) What circuit analysis technique must be used to find the steady-state expression for $v_o(t)$?
- b) Find the steady-state expression for $v_o(t)$.

Figure P9.68



- 10.39** a) Find the average power dissipated in each resistor in the circuit in Fig. P10.39.
 b) Check your answer by showing that the total power developed equals the total power absorbed.

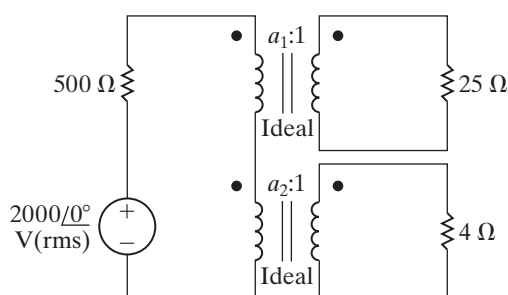
Figure P10.39



- 10.40** The sinusoidal voltage source in the circuit in Fig. P10.40 is developing an rms voltage of 2000 V. The 4 Ω load in the circuit is absorbing four times as much average power as the 25 Ω load. The two loads are matched to the sinusoidal source that has an internal impedance of $500\angle 0^\circ \Omega$.

- a) Specify the numerical values of a_1 and a_2 .
 b) Calculate the power delivered to the 25 Ω load.
 c) Calculate the rms value of the voltage across the 4 Ω resistor.

Figure P10.40



Section 10.6

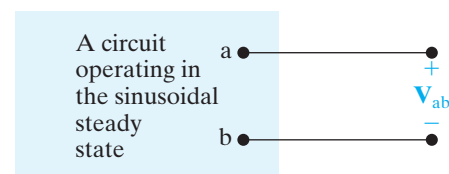
- 10.41** Suppose an impedance equal to the conjugate of the Thévenin impedance is connected to the terminals c, d of the circuit shown in Fig. P9.75.

- a) Find the average power developed by the sinusoidal voltage source.
 b) What percentage of the power developed by the source is lost in the linear transformer?

- 10.42** The phasor voltage $\mathbf{V}_{ab}$ in the circuit shown in Fig. P10.42 is $300\angle 0^\circ \text{ V(rms)}$ when no external load is connected to the terminals a, b. When a load having an impedance of $200 - j500 \Omega$ is connected across a, b, the value of $\mathbf{V}_{ab}$ is $156 - j42 \text{ V(rms)}$.

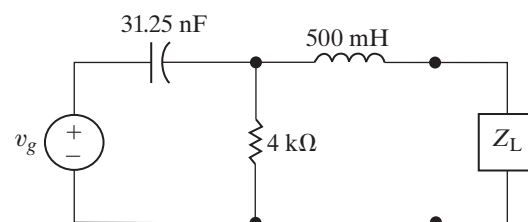
- a) Find the impedance that should be connected across a, b for maximum average power transfer.
 b) Find the maximum average power transferred to the load of (a).
 c) Construct the impedance of part (a) using components from Appendix H if the source frequency is 500 Hz.

Figure P10.42



- 10.43** a) Determine the load impedance for the circuit shown in Fig. P10.43 that will result in maximum average power being transferred to the load if $\omega = 8 \text{ krad/s}$.
 b) Determine the maximum average power delivered to the load from part (a) if $v_g = 10 \cos 8000t \text{ V}$.
 c) Repeat part (a) when Z_L consists of two components from Appendix H whose values yield a maximum average power closest to the value calculated in part (b).

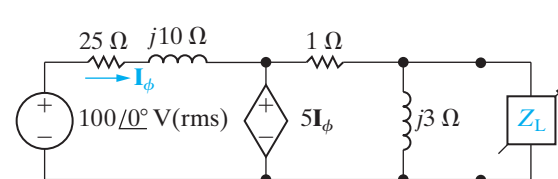
Figure P10.43



- 10.44** The load impedance Z_L for the circuit shown in Fig. P10.44 is adjusted until maximum average power is delivered to Z_L .

- a) Find the maximum average power delivered to Z_L .
 b) What percentage of the total power developed in the circuit is delivered to Z_L ?

Figure P10.44



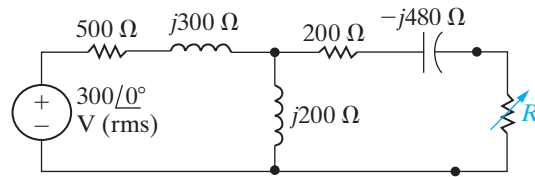
- 10.45** Prove that if only the magnitude of the load impedance can be varied, the most average power is

transferred to the load when $|Z_L| = |Z_{Th}|$. (Hint: In deriving the expression for the load's average power, write the load impedance (Z_L) in the form $Z_L = |Z_L|\cos\theta + j|Z_L|\sin\theta$, and note that only $|Z_L|$ is variable.)

- 10.46** The variable resistor in the circuit shown in Fig. P10.46 is adjusted until the average power it absorbs is maximum.

- Find R .
- Find the maximum average power.
- Find a resistor in Appendix H that would have the most average power delivered to it.

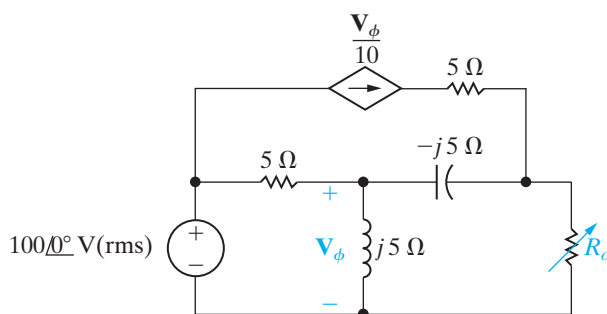
Figure P10.46



- 10.47** The variable resistor R_o in the circuit shown in Fig. P10.47 is adjusted until maximum average power is delivered to R_o .

- What is the value of R_o in ohms?
- Calculate the average power delivered to R_o .
- If R_o is replaced with a variable impedance Z_o , what is the maximum average power that can be delivered to Z_o ?
- In (c), what percentage of the circuit's developed power is delivered to the load Z_o ?

Figure P10.47

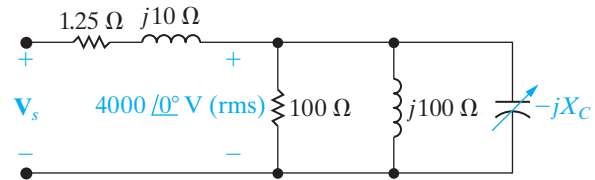


- 10.48** The sending-end voltage in the circuit seen in Fig. P10.48 is adjusted so that the rms value of the load voltage is always 4000 V. The variable capacitor is adjusted until the average power dissipated in the line resistance is minimum.

- If the frequency of the sinusoidal source is 60 Hz, what is the value of the capacitance in microfarads?
- If the capacitor is removed from the circuit, what percentage increase in the magnitude of $\mathbf{V}_s$ is necessary to maintain 4000 V at the load?

- If the capacitor is removed from the circuit, what is the percentage increase in line loss?

Figure P10.48

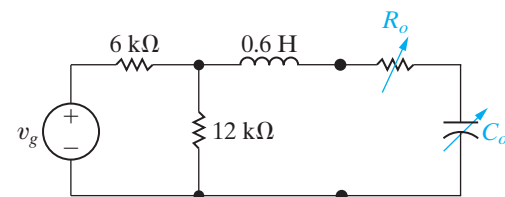


- 10.49** The peak amplitude of the sinusoidal voltage source in the circuit shown in Fig. P10.49 is 180 V, and its frequency is 5000 rad/s. The load resistor can be varied from 0 to 4000 Ω, and the load capacitor can be varied from 0.1 μF to 0.5 μF.

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- Calculate the average power delivered to the load when $R_o = 2000\ \Omega$ and $C_o = 0.2\ \mu\text{F}$.
- Determine the settings of R_o and C_o that will result in the most average power being transferred to R_o .
- What is the average power in (b)? Is it greater than the power in (a)?
- If there are no constraints on R_o and C_o , what is the maximum average power that can be delivered to a load?
- What are the values of R_o and C_o for the condition of (d)?
- Is the average power calculated in (d) larger than that calculated in (c)?

Figure P10.49



- 10.50** Assume that R_o in Fig. P10.49 can be varied between 0 and 10 kΩ. Repeat (b) and (c) of Problem 10.49.

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- Is the new average power calculated in (a) greater than that found in Problem 10.49(a)?
- Is the new average power calculated in (a) less than that found in 10.49(d)?

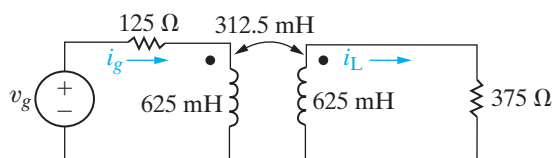
- 10.51** Find the steady-state expression for the currents i_g and i_L in the circuit in Fig. P10.51 when $v_g = 400\cos 400t\ \text{V}$.

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- Find the coefficient of coupling.
- Find the energy stored in the magnetically coupled coils at $t = 1.25\pi\ \text{ms}$ and $t = 2.5\pi\ \text{ms}$.

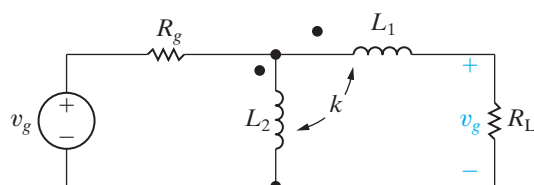
- d) Find the power delivered to the $375\ \Omega$ resistor.
- e) If the $375\ \Omega$ resistor is replaced by a variable resistor R_L , what value of R_L will yield maximum average power transfer to R_L ?
- f) What is the maximum average power in (e)?
- g) Assume the $375\ \Omega$ resistor is replaced by a variable impedance Z_L . What value of Z_L will result in maximum average power transfer to Z_L ?
- h) What is the maximum average power in (g)?

Figure P10.51



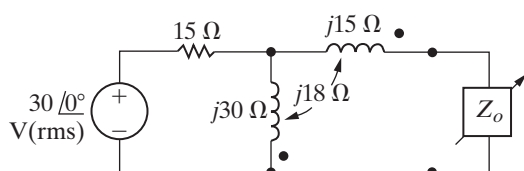
- 10.52** The value of the parameters in the circuit shown in Fig. P10.52 are $L_1 = 4\text{ mH}$; $L_2 = 1\text{ mH}$; $k = 0.75$; $R_g = 2\ \Omega$; and $R_L = 14\ \Omega$. If $v_g = 54\sqrt{2} \cos 1000t\text{ V}$, find
- a) the rms magnitude of v_o ;
 - b) the average power delivered to R_L ;
 - c) the percentage of the average power generated by the ideal voltage source that is delivered to R_L .

Figure P10.52



- 10.53** Assume the coefficient of coupling in the circuit in Fig. P10.52 is adjustable.
- a) Find the value of k that makes v_o equal to zero.
 - b) Find the power developed by the source when k has the value found in (a).
- 10.54** Assume the load resistor (R_L) in the circuit in Fig. P10.52 is adjustable.
- a) What value of R_L will result in the maximum average power being transferred to R_L ?
 - b) What is the value of the maximum power transferred?
- 10.55** Find the impedance seen by the ideal voltage source in the circuit in Fig. P10.55 when Z_o is adjusted for maximum average power transfer to Z_o .

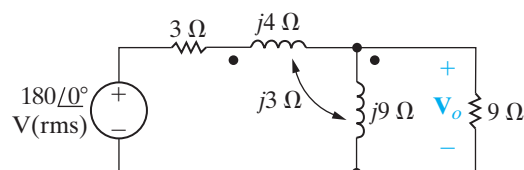
Figure P10.55



- 10.56** For the frequency-domain circuit in Fig. P10.56, calculate:

- a) the rms magnitude of V_o ;
- b) the average power dissipated in the $9\ \Omega$ resistor;
- c) the percentage of the average power generated by the ideal voltage source that is delivered to the $9\ \Omega$ load resistor.

Figure P10.56

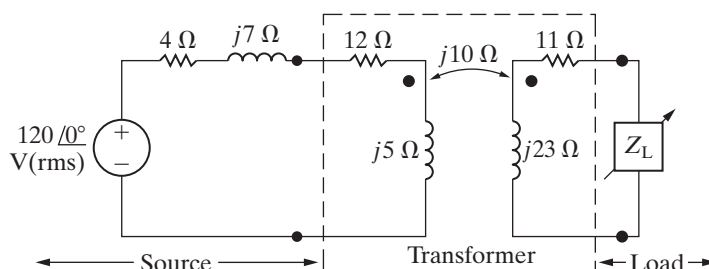


- 10.57** The $9\ \Omega$ resistor in the circuit in Fig. P10.56 is replaced with a variable impedance Z_o . Assume Z_o is adjusted for maximum average power transfer to Z_o .
- a) What is the maximum average power that can be delivered to Z_o ?
 - b) What is the average power developed by the ideal voltage source when maximum average power is delivered to Z_o ?
 - c) Choose single components from Appendix H to form an impedance that dissipates average power closest to the value in part (a). Assume the source frequency is 60 Hz .

- 10.58** The impedance Z_L in the circuit in Fig. P10.58 is adjusted for maximum average power transfer to Z_L . The internal impedance of the sinusoidal voltage source is $4 + j7\ \Omega$.

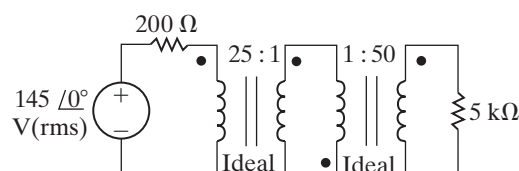
- a) What is the maximum average power delivered to Z_L ?
- b) What percentage of the average power delivered to the linear transformer is delivered to Z_L ?

Figure P10.58



- 10.59** Find the average power delivered to the $5\text{ k}\Omega$ resistor in the circuit of Fig. P10.59.

Figure P10.59



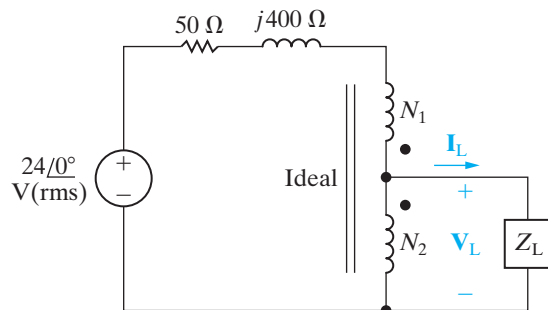
10.60 The ideal transformer connected to the $5\text{ k}\Omega$ load in Problem 10.59 is replaced with an ideal transformer that has a turns ratio of $1:a$.

- What value of a results in maximum average power being delivered to the $5\text{ k}\Omega$ resistor?
- What is the maximum average power?

10.61 The load impedance Z_L in the circuit in Fig. P10.61 is adjusted until maximum average power is transferred to Z_L .

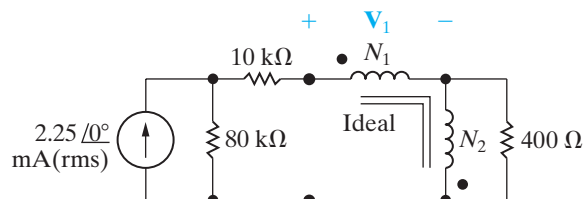
- Specify the value of Z_L if $N_1 = 3600$ turns and $N_2 = 600$ turns.
- Specify the values of $\mathbf{I}_L$ and $\mathbf{V}_L$ when Z_L is absorbing maximum average power.

Figure P10.61



- Find the turns ratio N_1/N_2 for the ideal transformer in the circuit in Fig. P10.62 so that maximum average power is delivered to the $400\text{ }\Omega$ load.
- Find the average power delivered to the $400\text{ }\Omega$ load.
- Find the voltage $\mathbf{V}_1$.
- What percentage of the power developed by the ideal current source is delivered to the $400\text{ }\Omega$ resistor?

Figure P10.62

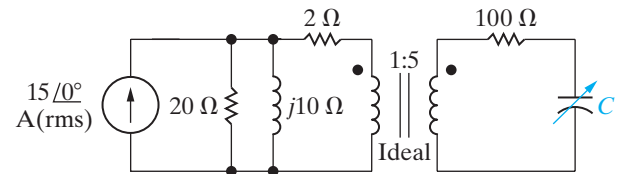


10.63 The sinusoidal voltage source in the circuit in Fig. P10.63 is operating at a frequency of 20 krad/s . The variable capacitive reactance in the circuit is adjusted until the average power delivered to the $100\text{ }\Omega$ resistor is as large as possible.

- Find the value of C in microfarads.
- When C has the value found in (a), what is the average power delivered to the $100\text{ }\Omega$ resistor?
- Replace the $100\text{ }\Omega$ resistor with a variable resistor R_o . Specify the value of R_o so that maximum average power is delivered to R_o .

- What is the maximum average power that can be delivered to R_o ?

Figure P10.63

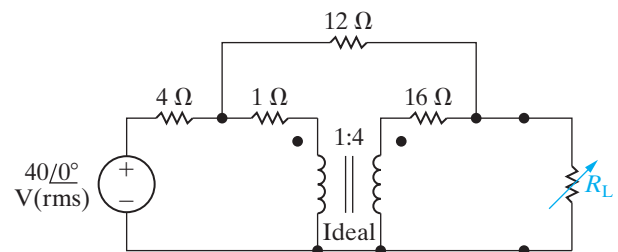


10.64 The variable load resistor R_L in the circuit shown in Fig. P10.64 is adjusted for maximum average power transfer to R_L .

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- Find the maximum average power.
- What percentage of the average power developed by the ideal voltage source is delivered to R_L when R_L is absorbing maximum average power?
- Test your solution by showing that the power developed by the ideal voltage source equals the power dissipated in the circuit.

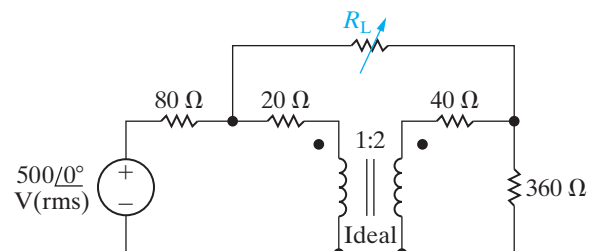
Figure P10.64



10.65 Repeat Problem 10.64 for the circuit shown in Fig. P10.65.

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Figure P10.65



10.66 a) If N_1 equals 1000 turns, how many turns should be placed on the N_2 winding of the ideal transformer in the circuit seen in Fig. P10.66 so that maximum average power is delivered to the $6800\text{ }\Omega$ load?

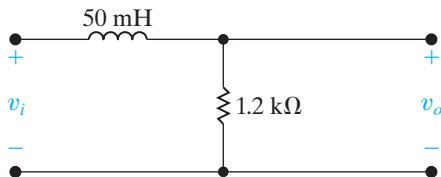
- Find the average power delivered to the $6800\text{ }\Omega$ resistor.
- What percentage of the average power delivered by the ideal voltage source is dissipated in the linear transformer?

Problems

Section 14.2

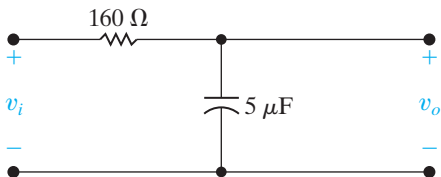
- 14.1** a) Find the cutoff frequency in hertz for the RL filter shown in Fig. P14.1.
 b) Calculate $H(j\omega)$ at ω_c , $0.125\omega_c$, and $8\omega_c$.
 c) If $v_i = 20 \cos \omega t$ V, write the steady-state expression for v_o when $\omega = \omega_c$, $\omega = 0.125\omega_c$, and $\omega = 8\omega_c$.

Figure P14.1



- 14.2** a) Find the cutoff frequency (in hertz) of the low-pass filter shown in Fig. P14.2.
 b) Calculate $H(j\omega)$ at ω_c , $0.1\omega_c$, and $10\omega_c$.
 c) If $v_i = 25 \cos \omega t$ mV, write the steady-state expression for v_o when $\omega = \omega_c$, $0.1\omega_c$, and $10\omega_c$.

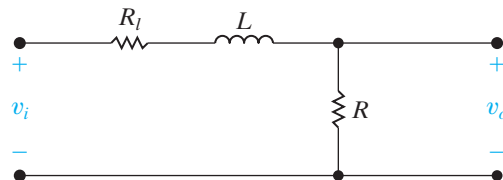
Figure P14.2



- 14.3** A resistor, denoted as R_I , is added in series with the inductor in the circuit in Fig. 14.4(a). The new low-pass filter circuit is shown in Fig. P14.3.

- a) Derive the expression for $H(s)$ where $H(s) = V_o/V_i$.
 b) At what frequency will the magnitude of $H(j\omega)$ be maximum?
 c) What is the maximum value of the magnitude of $H(j\omega)$?
 d) At what frequency will the magnitude of $H(j\omega)$ equal its maximum value divided by $\sqrt{2}$?
 e) Assume a resistance of 300Ω is added in series with the 50 mH inductor in the circuit in Fig. P14.1. Find ω_c , $H(j0)$, $H(j\omega_c)$, $H(j0.2\omega_c)$, and $H(j5\omega_c)$.

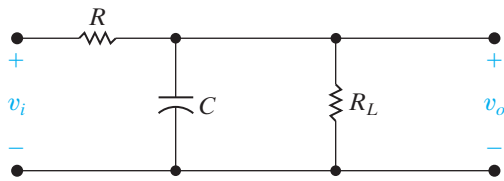
Figure P14.3



- 14.4** A resistor denoted as R_L is connected in parallel with the capacitor in the circuit in Fig. 14.7. The loaded low-pass filter circuit is shown in Fig. P14.7.

- Derive the expression for the voltage transfer function V_o/V_i .
- At what frequency will the magnitude of $H(j\omega)$ be maximum?
- What is the maximum value of the magnitude of $H(j\omega)$?
- At what frequency will the magnitude of $H(j\omega)$ equal its maximum value divided by $\sqrt{2}$?
- Assume a resistance of $320\ \Omega$ is added in parallel with the $5\ \mu\text{F}$ capacitor in the circuit in Fig. P14.4. Find ω_c , $H(j0)$, $H(j\omega_c)$, $H(j0.2\omega_c)$, and $H(j5\omega_c)$.

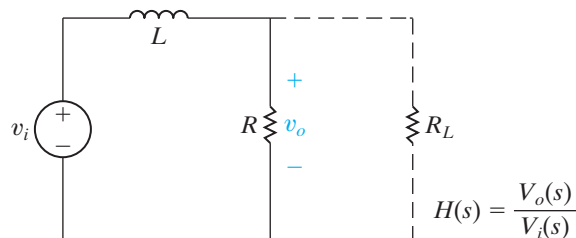
Figure P14.4



14.5 Consider the low-pass filter in Fig. P14.5, which has a load resistor R_L .

- What is the transfer function of the unloaded filter?
- What is the transfer function of the loaded filter?
- Compare the transfer function of the unloaded filter in part (a) and the transfer function of the loaded filter in part (b). Are the cutoff frequencies different? Are the passband gains different?
- If $R = 270\ \Omega$ and $L = 5\ \text{mH}$, what is the cutoff frequency of the unloaded filter in rad/s?
- What is the smallest value of load resistance that can be used with the filter components in part (d) so that the cutoff frequency of the resulting filter is no more than 2% different from the unloaded filter?

Figure P14.5



14.6 Consider the low-pass filter designed in Problem 14.5.

- Assume the cutoff frequency cannot decrease by more than 10% from the specified value, 8 krad/s. What is the smallest value of load resistance that can be connected across the output terminals of the filter?

- If the resistor found in part (a) is connected across the output terminals of the filter, what is the magnitude of $H(j\omega)$ when $\omega = 0$?

14.7 Use a 1 mH inductor to design a low-pass, RL , passive filter with a cutoff frequency of 5 kHz.

DESIGN
PROBLEM
PSPICE
MULTISIM

- Specify the value of the resistor.
- A load having a resistance of $68\ \Omega$ is connected across the output terminals of the filter. What is the corner, or cutoff, frequency of the loaded filter in hertz?
- If you must use a single resistor from Appendix H for part (a), what resistor should you use? What is the resulting cutoff frequency of the filter?

14.8 Use a 10 mH inductor to design a low-pass passive filter with a cutoff frequency of 1600 rad/s.

- Specify the cutoff frequency in hertz.
- Specify the value of the filter resistor.
- Assume the cutoff frequency cannot decrease by more than 10%. What is the smallest value of load resistance that can be connected across the output terminals of the filter?
- If the resistor found in (c) is connected across the output terminals, what is the magnitude of $H(j\omega)$ when $\omega = 0$?

14.9 Design a passive RC low pass filter (see Fig. 14.7) with a cutoff frequency of 100 Hz using a $4.7\ \mu\text{F}$ capacitor.

- What is the cutoff frequency in rad/s?
- What is the value of the resistor?
- Draw your circuit, labeling the component values and output voltage.
- What is the transfer function of the filter in part (c)?
- If the filter in part (c) is loaded with a resistor whose value is the same as the resistor part (b), what is the transfer function of this loaded filter?
- What is the cutoff frequency of the loaded filter from part (e)?
- What is the gain in the pass band of the loaded filter from part (e)?

14.10 Use a 500 nF capacitor to design a low-pass passive filter with a cutoff frequency of 50 krad/s.

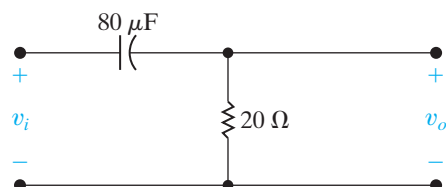
DESIGN
PROBLEM
PSPICE
MULTISIM

- Specify the cutoff frequency in hertz.
- Specify the value of the filter resistor.
- Assume the cutoff frequency cannot increase by more than 5%. What is the smallest value of load resistance that can be connected across the output terminals of the filter?
- If the resistor found in (c) is connected across the output terminals, what is the magnitude of $H(j\omega)$ when $\omega = 0$?

Section 14.3

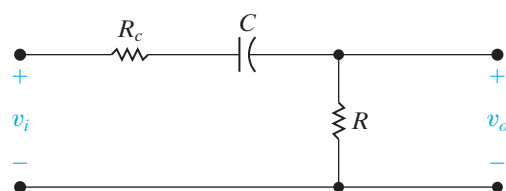
- 14.11** a) Find the cutoff frequency (in hertz) for the high-pass filter shown in Fig. P14.11.
 b) Find $H(j\omega)$ at ω_c , $0.125\omega_c$, and $8\omega_c$.
 c) If $v_i = 75 \cos \omega t$ V, write the steady-state expression for v_o when $\omega = \omega_c$, $\omega = 0.125\omega_c$, and $\omega = 8\omega_c$.

Figure P14.11



- 14.12** A resistor, denoted as R_c , is connected in series with the capacitor in the circuit in Fig. 14.11(a). The new high-pass filter circuit is shown in Fig. P14.12.
- Derive the expression for $H(s)$ where $H(s) = V_o(s)/V_i(s)$.
 - At what frequency will the magnitude of $H(j\omega)$ be maximum?
 - What is the maximum value of the magnitude of $H(j\omega)$?
 - At what frequency will the magnitude of $H(j\omega)$ equal its maximum value divided by $\sqrt{2}$?
 - Assume a resistance of 5Ω is connected in series with the $80 \mu\text{F}$ capacitor in the circuit in Fig. P14.11. Calculate ω_c , $H(j\omega_c)$, $H(j0.125\omega_c)$, and $H(j8\omega_c)$.

Figure P14.12



- 14.13** Using a 100 nF capacitor, design a high-pass passive filter with a cutoff frequency of 300 Hz .

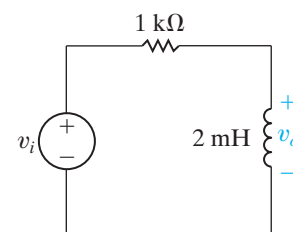
DESIGN
PROBLEM

- Specify the value of R in kilohms.
- A $47 \text{ k}\Omega$ resistor is connected across the output terminals of the filter. What is the cutoff frequency, in hertz, of the loaded filter?

- 14.14** Consider the circuit shown in Fig. P14.14.

- What is the transfer function, $H(s) = V_o(s)/V_i(s)$, of this filter?
- What is the cutoff frequency of this filter?
- What is the magnitude of the filter's transfer function at $s = j\omega_c$?

Figure P14.14



- 14.15** Suppose a 150Ω load resistor is attached to the filter in Fig. P14.11.

- What is the transfer function, $H(s) = V_o(s)/V_i(s)$, of this filter?
- What is the cutoff frequency of this filter?
- How does the cutoff frequency of the loaded filter compare with the cutoff frequency of the unloaded filter in Fig. P14.11?
- What else is different for these two filters?

- 14.16** Design a passive RC high pass filter (see Fig. 14.10[a]) with a cutoff frequency of 5 kHz using a 10 pF capacitor.

- What is the cutoff frequency in rad/s?
- What is the value of the resistor?
- Draw your circuit, labeling the component values and output voltage.
- What is the transfer function of the filter in part (c)?
- If the filter in part (c) is loaded with a resistor whose value is the same as the resistor in (b), what is the transfer function of this loaded filter?
- What is the cutoff frequency of the loaded filter from part (e)?
- What is the gain in the pass band of the loaded filter from part (e)?

- 14.17** Using a $100 \mu\text{H}$ inductor, design a high-pass, RL , passive filter with a cutoff frequency of 1500 rad/s .

DESIGN
PROBLEMPSPICE
MULTISIM

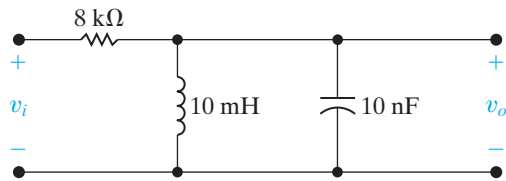
- Specify the value of the resistance, selecting from the components in Appendix H.
- Assume the filter is connected to a pure resistive load. The cutoff frequency is not to drop below 1200 rad/s . What is the smallest load resistor from Appendix H that can be connected across the output terminals of the filter?

Section 14.4

- 14.18** For the bandpass filter shown in Fig. P14.18, find (a) ω_o , (b) f_o , (c) Q , (d) ω_{c1} , (e) f_{c1} , (f) ω_{c2} , (g) f_{c2} , and (h) β .

PSPICE
MULTISIM

Figure P14.18



- 14.19** Calculate the center frequency, the bandwidth, and the quality factor of a bandpass filter that has an upper cutoff frequency of 141 krad/s and a lower cutoff frequency of 133 krad/s.
- 14.20** A bandpass filter has a center, or resonant, frequency of 50 krad/s and a quality factor of 4. Find the bandwidth, the upper cutoff frequency, and the lower cutoff frequency. Express all answers in kilohertz.
- 14.21** Design a series RLC bandpass filter (see Fig. 14.19[a]) with a quality of 8 and a center frequency of 50 krad/s, using a $0.01 \mu\text{F}$ capacitor.
- Draw your circuit, labeling the component values and output voltage.
 - For the filter in part (a), calculate the bandwidth and the values of the two cutoff frequencies.
- 14.22** The input to the series RLC bandpass filter designed in Problem 14.21 is $5\cos\omega t$ V. Find the voltage drop across the resistor when (a) $\omega = \omega_o$; (b) $\omega = \omega_{c1}$; (c) $\omega = \omega_{c2}$; (d) $\omega = 0.1\omega_o$; (e) $\omega = 10\omega_o$.
- 14.23** The input to the series RLC bandpass filter designed in Problem 14.21 is $5\cos\omega t$ V. Find the voltage drop across the series combination of the inductor and capacitor when (a) $\omega = \omega_o$; (b) $\omega = \omega_{c1}$; (c) $\omega = \omega_{c2}$; (d) $\omega = 0.1\omega_o$; (e) $\omega = 10\omega_o$.
- 14.24** a) Using the relationship between the bandwidth and the cutoff frequencies and the relationship between the center frequency and the cutoff frequencies, show that

$$\omega_{c1} = -\frac{\beta}{2} + \sqrt{\left(\frac{\beta}{2}\right)^2 + \omega_o^2};$$

$$\omega_{c2} = \frac{\beta}{2} + \sqrt{\left(\frac{\beta}{2}\right)^2 + \omega_o^2}.$$

- b) Using the expressions for the two cutoff frequencies in terms of the center frequency and the bandwidth given in part (a), show that

$$\omega_{c1} = \omega_o \cdot \left[-\frac{1}{2Q} + \sqrt{1 + \left(\frac{1}{2Q}\right)^2} \right],$$

$$\omega_{c2} = \omega_o \cdot \left[\frac{1}{2Q} + \sqrt{1 + \left(\frac{1}{2Q}\right)^2} \right].$$

- 14.25** Using a 50 nF capacitor in the bandpass circuit shown in Fig. 14.22, design a filter with a quality factor of 5 and a center frequency of 20 krad/s.

DESIGN
PROBLEM
PSPICE
MULTISIM

- Specify the numerical values of R and L .
- Calculate the upper and lower cutoff frequencies in kilohertz.
- Calculate the bandwidth in hertz.

- 14.26** Design a series RLC bandpass filter using only three components from Appendix H that comes closest to meeting the filter specifications in Problem 14.26.

- Draw your filter, labeling all component values and the input and output voltages.
- Calculate the percent error in this new filter's center frequency and quality factor when compared to the values specified in Problem 14.26.

- 14.27** Use a 5 nF capacitor to design a series RLC bandpass filter, as shown at the top of Fig. 14.27. The center frequency of the filter is 8 kHz, and the quality factor is 2.

DESIGN
PROBLEM
PSPICE
MULTISIM

- Specify the values of R and L .
- What is the lower cutoff frequency in kilohertz?
- What is the upper cutoff frequency in kilohertz?
- What is the bandwidth of the filter in kilohertz?

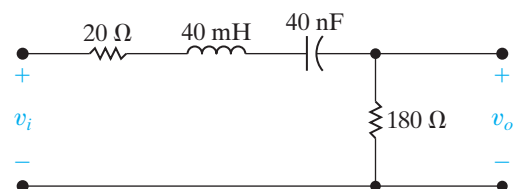
- 14.28** Design a series RLC bandpass filter using only three components from Appendix H that comes closest to meeting the filter specifications in Problem 14.27.

- Draw your filter, labeling all component values and the input and output voltages.
- Calculate the percent error in this new filter's center frequency and quality factor when compared to the values specified in Problem 14.27.

- 14.29** For the bandpass filter shown in Fig. P14.29, calculate the following: (a) f_o ; (b) Q ; (c) f_{c1} ; (d) f_{c2} ; and (e) β .

PSPICE
MULTISIM

Figure P14.29



- 14.30** The input voltage in the circuit in Fig. P14.29 is $10\cos\omega t$ V. Calculate the output voltage when (a) $\omega = \omega_o$; (b) $\omega = \omega_{c1}$; and (c) $\omega = \omega_{c2}$.

- 14.31** Consider the circuit shown in Fig. P14.31.

PSPICE
MULTISIM

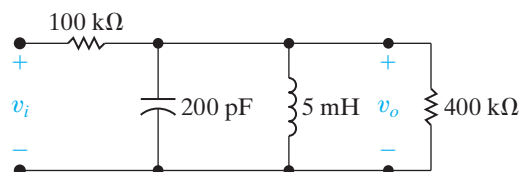
- Find ω_o .
- Find β .
- Find Q .

- d) Find the steady-state expression for v_o when $v_i = 250 \cos \omega_o t$ mV.
- e) Show that if R_L is expressed in kilohms the Q of the circuit in Fig. P14.31 is

$$Q = \frac{20}{1 + 100/R_L}.$$

- f) Plot Q versus R_L for $20 \text{ k}\Omega \leq R_L \leq 2 \text{ M}\Omega$.

Figure P14.31



- 14.32** A block diagram of a system consisting of a sinusoidal voltage source, a series RLC bandpass filter, and a load is shown in Fig. P14.32. The internal impedance of the sinusoidal source is $80 + j0 \Omega$, and the impedance of the load is $480 + j0 \Omega$. The RLC series bandpass filter has a 20 nF capacitor, a center frequency of 50 krad/s, and a quality factor of 6.25.

- Draw a circuit diagram of the system.
- Specify the numerical values of L and R for the filter section of the system.
- What is the quality factor of the interconnected system?
- What is the bandwidth (in hertz) of the interconnected system?

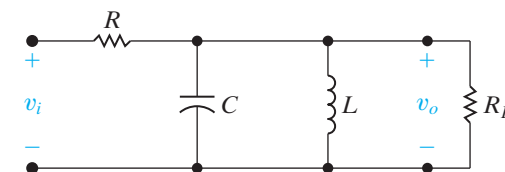
Figure P14.32



- 14.33** The purpose of this problem is to investigate how a resistive load connected across the output terminals of the bandpass filter shown in Fig. 14.19 affects the quality factor and hence the bandwidth of the filtering system. The loaded filter circuit is shown in Fig. P14.33.
- Calculate the transfer function V_o/V_i for the circuit shown in Fig. P14.33.
 - What is the expression for the bandwidth of the system?
 - What is the expression for the loaded bandwidth (β_L) as a function of the unloaded bandwidth (β_U)?

- What is the expression for the quality factor of the system?
- What is the expression for the loaded quality factor (Q_L) as a function of the unloaded quality factor (Q_U)?
- What are the expressions for the cutoff frequencies ω_{c1} and ω_{c2} ?

Figure P14.33

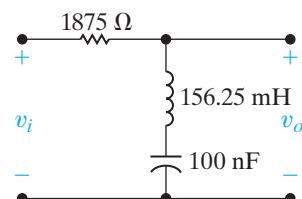


- 14.34** The parameters in the circuit in Fig. P14.33 are $R = 2.4 \text{ k}\Omega$, $C = 50 \text{ pF}$, and $L = 2 \mu\text{H}$. The quality factor of the circuit is not to drop below 7.5. What is the smallest permissible value of the load resistor R_L ?

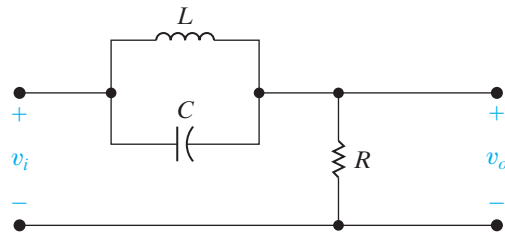
Section 14.5

- 14.35** For the bandreject filter in Fig. P14.35, calculate (a) ω_o ; (b) f_o ; (c) Q ; (d) β in hertz; (e) ω_{c1} ; (f) f_{c1} ; (g) ω_{c2} ; and (h) f_{c2} .

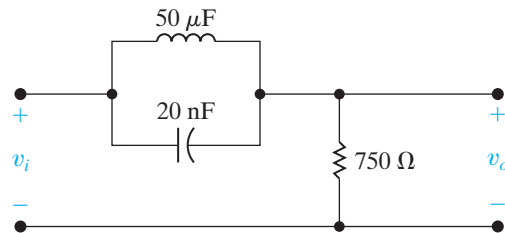
Figure P14.35



- 14.36** For the bandreject filter in Fig. P14.35,
- Find $H(j\omega)$ at ω_o , ω_{c1} , ω_{c2} , $0.1\omega_o$, and $10\omega_o$.
 - If $v_i = 80 \cos \omega t$ V, write the steady-state expression for v_o when $\omega = \omega_o$, $\omega = \omega_{c1}$, $\omega = \omega_{c2}$, $\omega = 0.1\omega_o$, and $\omega = 10\omega_o$.
- 14.37**
- Show (via a qualitative analysis) that the circuit in Fig. P14.37 is a bandreject filter.
 - Support the qualitative analysis of (a) by finding the voltage transfer function of the filter.
 - Derive the expression for the center frequency of the filter.
 - Derive the expressions for the cutoff frequencies ω_{c1} and ω_{c2} .
 - What is the expression for the bandwidth of the filter?
 - What is the expression for the quality factor of the circuit?

Figure P14.37

- 14.38** For the bandreject filter in Fig. P14.38, calculate (a) ω_o ; (b) f_o ; (c) Q ; (d) ω_{c1} ; (e) f_{c1} ; (f) ω_{c2} ; (g) f_{c2} ; and (h) β in kilohertz.

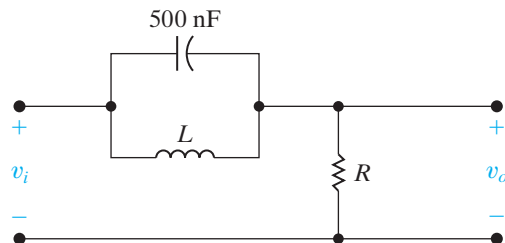
PSPICE
MULTISIM**Figure P14.38**

- 14.39** Design an RLC bandreject filter (see Fig. 14.28[a]) with a quality of 2.5 and a center frequency of 25 krad/s, using a 200 nF capacitor.
- Draw your circuit, labeling the component values and output voltage.
 - For the filter in part (a), calculate the bandwidth and the values of the two cutoff frequencies.
- 14.40** The input to the RLC bandreject filter designed in Problem 14.39 is $10\cos\omega t$ V. Find the voltage drop across the series combination of the inductor and capacitor when (a) $\omega = \omega_o$; (b) $\omega = \omega_{c1}$; (c) $\omega = \omega_{c2}$; (d) $\omega = 0.125\omega_o$; (e) $\omega = 8\omega_o$.
- 14.41** The input to the RLC bandreject filter designed in Problem 14.39 is $10\cos\omega t$ V. Find the voltage drop across the resistor when (a) $\omega = \omega_o$; (b) $\omega = \omega_{c1}$; (c) $\omega = \omega_{c2}$; (d) $\omega = 0.125\omega_o$; (e) $\omega = 8\omega_o$.

- 14.42** Use a 500 nF capacitor to design a bandreject filter, as shown in Fig. P14.42. The filter has a center frequency of 4 kHz and a quality factor of 5.

DESIGN
PROBLEMPSPICE
MULTISIM

- Specify the numerical values of R and L .
- Calculate the upper and lower corner, or cutoff, frequencies in kilohertz.
- Calculate the filter bandwidth in kilohertz.

Figure P14.42

- 14.43** Assume the bandreject filter in Problem 14.42 is loaded with a $1\text{ k}\Omega$ resistor.

PSPICE
MULTISIM

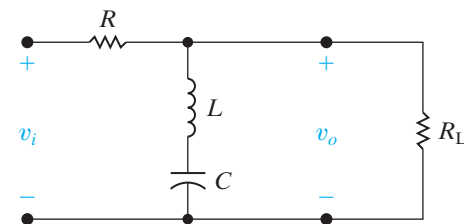
- What is the quality factor of the loaded circuit?
- What is the bandwidth (in kilohertz) of the loaded circuit?
- What is the upper cutoff frequency in kilohertz?
- What is the lower cutoff frequency in kilohertz?

- 14.44** Design a parallel RLC bandreject filter using only three components from Appendix H that comes closest to meeting the filter specifications in Problem 14.42.

- Draw your filter, labeling all component values and the input and output voltages.
- Calculate the percent error in this new filter's center frequency and quality factor when compared to the values specified in Problem 14.42.

- 14.45** The purpose of this problem is to investigate how a resistive load connected across the output terminals of the bandreject filter shown in Fig. 14.28(a) affects the behavior of the filter. The loaded filter circuit is shown in Fig. P14.45.

- Find the voltage transfer function V_o/V_i .
- What is the expression for the center frequency?
- What is the expression for the bandwidth?
- What is the expression for the quality factor?
- Evaluate $H(j\omega_o)$.
- Evaluate $H(j0)$.
- Evaluate $H(j\infty)$.
- What are the expressions for the corner frequencies ω_{c1} and ω_{c2} ?

Figure P14.45

- 14.46** The parameters in the circuit in Fig. P14.45 are $R = 30\ \Omega$, $L = 1\ \mu\text{H}$, $C = 4\ \text{pF}$, and $R_L = 150\ \Omega$.

PSPICE
MULTISIM

- Find ω_o , β (in kilohertz), and Q .
- Find $H(j0)$ and $H(j\infty)$.
- Find f_{c2} and f_{c1} .
- Show that if R_L is expressed in ohms the Q of the circuit is

$$Q = \frac{50}{3} [1 + (30/R_L)].$$

- Plot Q versus R_L for $10\ \Omega \leq R_L \leq 300\ \Omega$.

14.47 A $500\ \Omega$ load is added to the bandreject filter shown in Fig. P14.42. The center frequency of the filter is 25 krad/s , and the capacitor is 25 nF . At very low and very high frequencies, the amplitude of the sinusoidal output voltage should be at least 90% of the amplitude of the sinusoidal input voltage.

PSPICE
MULTISIM

- Specify the numerical values of R and L .
- What is the quality factor of the circuit?

Sections 14.1–14.5

14.48 Given the following voltage transfer function:

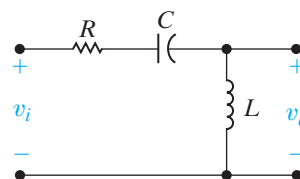
$$H(s) = \frac{V_o}{V_i} = \frac{25 \times 10^6}{s^2 + 1000s + 25 \times 10^6}$$

- At what frequencies (in radians per second) is the magnitude of the transfer function equal to unity?
- At what frequency is the magnitude of the transfer function maximum?
- What is the maximum value of the transfer function magnitude?

14.49 Consider the series RLC circuit shown in Fig. P14.49. When the output is the voltage across the resistor, we know this circuit is a bandpass filter. When the output is the voltage across the series combination of the inductor and capacitor, we know this circuit is a bandreject filter. This problem investigates the behavior of this circuit when the output is across the inductor.

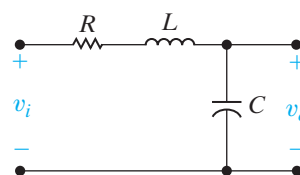
- Find the transfer function, $H(s) = V_o(s)/V_i(s)$ when $V_o(s)$ is the voltage across the inductor.
- Find the magnitude of the transfer function in part (a) for very low frequencies.
- Find the magnitude of the transfer function in part (a) for very high frequencies.
- Based on your answers in parts (b) and (c), what type of filter is this?
- Suppose $R = 600\ \Omega$, $L = 400\text{ mH}$, $C = 2.5\ \mu\text{F}$. Calculate the cutoff frequency of this filter, that is, the frequency at which the magnitude of the transfer function is $1/\sqrt{2}$.

Figure P14.49



14.50 Repeat Problem 14.49 for the circuit shown in Fig. P14.50. Note that the output voltage is now the voltage across the capacitor.

Figure P14.50



14.51 Design a series RLC bandpass filter (see Fig. 14.27) for detecting the low-frequency tone generated by pushing a telephone button as shown in Fig. 14.32.

PRACTICAL
PERSPECTIVE
DESIGN
PROBLEM

- Calculate the values of L and C that place the cutoff frequencies at the edges of the DTMF low-frequency band. Note that the resistance in standard telephone circuits is always $R = 600\ \Omega$.
- What is the output amplitude of this circuit at each of the low-band frequencies, relative to the peak amplitude of the bandpass filter?
- What is the output amplitude of this circuit at the lowest of the high-band frequencies?

14.52 Design a DTMF high-band bandpass filter similar to the low-band filter design in Problem 14.51. Be sure to include the fourth high-frequency tone, 1633 Hz , in your design. What is the response amplitude of your filter to the highest of the low-frequency DTMF tones?

PRACTICAL
PERSPECTIVE
DESIGN
PROBLEM

14.53 The 20 Hz signal that rings a telephone's bell has to have a very large amplitude to produce a loud enough bell signal. How much larger can the ringing signal amplitude be, relative to the low-band DTMF signal, so that the response of the filter in Problem 14.51 is no more than half as large as the largest of the DTMF tones?

PRACTICAL
PERSPECTIVE
DESIGN
PROBLEM